DISSECTING THE SCORE: CALIBRATED DIAGNOSIS OF EEG FOUNDATION MODELS

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ABSTRACT

EEG foundation models (EEG-FMs) continue to achieve strong performance on downstream tasks, but task scores alone do not reveal which properties of the signal drive their predictions. In particular, it remains unclear whether predictions depend on physiologically meaningful features of EEG or on simpler properties such as electrode arrangement and signal amplitude. Input perturbations do not fully resolve this question, because their effects can also depend on the adapter, the readout, and artifacts introduced by the perturbation itself. We therefore propose a cross-architecture diagnostic benchmark that uses neurophysiology as a common reference. For each model, we first fix the adapter, pretrained backbone, and readout, and then apply eight diagnostic protocols to the input without further training. Paired controls are used to separate the intended intervention from major side effects, and three representative protocols are validated on synthetic EEG with known signal dependencies. Across five released models and four tasks, we observe a recurring pattern: predictions are often more sensitive to global or low-level signal properties than to the specific brain regions and EEG rhythms expected to be relevant to the task. On motor imagery, swapping corresponding electrodes between the two hemispheres reduces AUROC by 0.18–0.28, whereas completely masking the predefined central sensorimotor region changes performance by less than 0.04. On seizure detection, two models that preserve input amplitude remain highly predictive even after the original waveform is fully replaced by RMS-matched background, yet fall below chance when the original waveform is kept but strongly attenuated. The same design also allows us to test where these behaviors can be changed. With the pretrained backbone kept frozen, a mirror-augmented readout refit makes the motor imagery predictions respond more consistently to left–right electrode swapping. This shows that some behaviors revealed by the diagnostic tests can be changed through the downstream interface without modifying the pretrained backbone. Overall, the benchmark provides a controlled way to uncover signal dependencies that are hidden by task scores and to test whether they can be modified at the downstream stage.

1 INTRODUCTION

EEG foundation models (EEG-FMs) are emerging rapidly (Zhou et al., 2025; Jiang et al., 2024; Yang et al., 2023; Wang et al., 2025; El Ouahidi et al., 2025), and most benchmarks compare task scores after attaching a downstream readout to a pretrained backbone (Xiong et al., 2026; Kastrati et al., 2025; Kontras et al., 2026; Lu et al., 2026). But task scores alone do not reveal which signal properties drive the predictions, and recent work has begun to examine this question. Linear probing on frozen representations (Kontras et al., 2026), robustness and spatial perturbation analyses (Lu et al., 2026; Shen et al., 2026), and various representation audits (Širca et al., 2026; Wang et al., 2026; Tang et al., 2026; Lin et al., 2026; Zare, 2026; Lehn-Schiøler et al., 2026; Yu et al., 2026) have been proposed, each with its own tool. However, these analyses face three obstacles in cross-model evaluation:

First, confounding. A score and its drop reflect the whole evaluation route, including the adapter, the readout, and the checkpoint rule, not the backbone alone. Replacing only the readout changes the score (Xiong et al., 2026; Širca et al., 2026), and can even reverse model rankings across benchmarks

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on the same datasets (Lahiri et al., 2026). Input perturbations also have side effects (masking removes channels, truncation adds zero padding, and a mixed-in background adds energy). **Second, lack of ground truth.** Real EEG offers no ground truth for which channel, band, or time window carries the label. **Third, lack of attribution.** Without fixing each component of the evaluation route, one cannot attribute a phenomenon to the interface (adapter or readout) or to the backbone, nor decide which of them to improve.

To address them, the benchmark is built on two-stage decoupling and checked by calibration on synthetic ground truth. In Stage I, we fix each model's adapter, backbone, and readout, fitting every readout with the same data, form, and checkpoint rule. In Stage II, we apply eight protocols to the raw input without further training and pair every operation that has a side effect with a control. The protocols are organized by three physical dimensions of the EEG signal, giving different architectures one frame of reference. Three representative protocols are first validated on synthetic EEG with known signal dependencies (Section 4.1). The readings of the protocols form the model's response profile.

Our contributions are three-fold:

1. **A controlled diagnostic benchmark with planted-ground-truth calibration.** We propose a cross-architecture benchmark that takes EEG neurophysiology as its frame of reference and combines two-stage decoupling, paired controls, and calibration on synthetic ground truth (Sections 3 and 4.1). It applies the same physical intervention before model-specific input processing, allowing heterogeneous models to be compared under a common experimental setup.
2. **Recurring patterns that scalar scores conceal.** The profiles of five released EEG-FMs across four tasks and eight protocols (Section 4) reveal one shared preference: reliance on coarse structure and weak response to the fine structure that physiology names. In space, predictions are highly sensitive to swapping corresponding electrodes between hemispheres, while regional masking has a much smaller effect. In the temporal–spectral dimension, predictions rely on the stationary content of a short window and respond weakly to the microstructure of physiological rhythms; in artifact and background, predictions rely on input amplitude and respond weakly to the waveform once amplitude is removed.
3. **A closed loop from diagnosis to improvement.** Decoupling turns attribution into an operable experiment: replacing only the interface and re-running the same protocol assigns a phenomenon to the interface or to the backbone (Section 5). At the interface, refitting the readout recovers properties the representation held unused, and the adapter's normalization decides whether the input amplitude reaches the backbone; deployment guidelines follow from both. A phenomenon rooted in the backbone becomes a named target for next-generation pretraining with a test that could reject it.

2 Related Work

2.1 EEG-FM Benchmarks and Concurrent Audits

Benchmarks of the same paradigm rank models by task score on their task sets (Jayaram & Barachant, 2018; Aristimunha et al., 2025; Xiong et al., 2026; Kastrati et al., 2025; Kontras et al., 2026; Lu et al., 2026; Liu et al., 2026; Banville et al., 2026; Shen et al., 2026). The whole evaluation route, however, is itself confounded: evaluation settings alone reverse rankings on the same datasets (Lahiri et al., 2026), channel adaptation interacts with the training regime (frozen or fine-tuned) (Kokate et al., 2026), and task definitions are not standardized (Qin et al., 2026), so comparing models is in part comparing routes.

Later benchmarks and concurrent audits therefore read the representation directly, in two ways. One reads inside the model, through a linear readout on the frozen backbone (Kontras et al., 2026), probing and erasure of concept and identity encoding (Tang et al., 2026; Lin et al., 2026), sparse feature dictionaries (Lehn-Schiøler et al., 2026), or frequency preference (Yu et al., 2026). The other varies the signal at the input and reads the prediction: temporal, spatial, and spectral surrogates (Wang et al., 2026), channel masking and additive noise (Širca et al., 2026), and aperiodic ablation (Lin et al., 2026). Checks on the reading also differ: a random-feature baseline (Tang et al., 2026), a null for subject variance (Lin et al., 2026), negative controls from permuted labels (Zare, 2026).

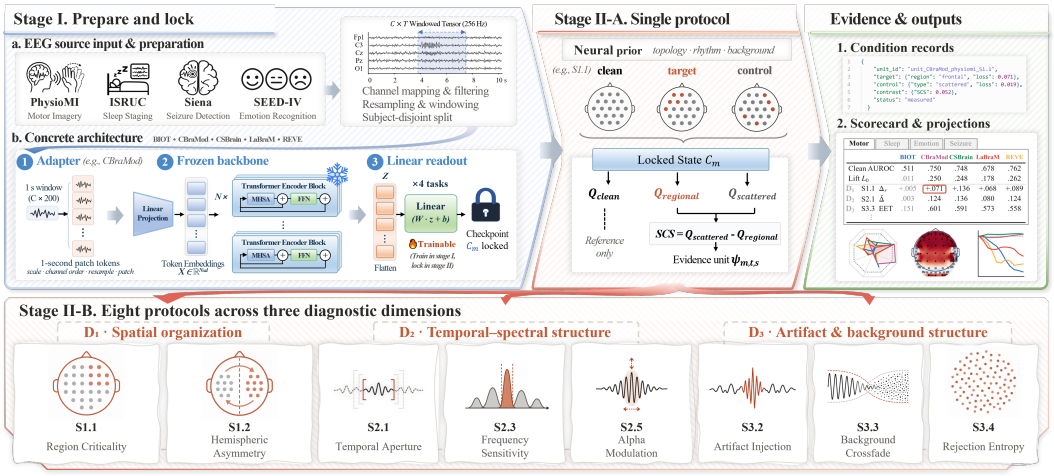


Figure 1: **Overview of the diagnostic pipeline.** Stage I trains and fixes one adapter–backbone–readout route per model. Stage II keeps that route fixed and applies eight pre-specified input perturbation protocols without further training. Synthetic calibration and interface refits are separate validation and follow-up analysis.

Each path has its trade-off: internal readings are specific to each architecture's representation space; input perturbations apply across models, but unless the route state is fixed, the reading changes with fine-tuning. This benchmark therefore fixes the state before perturbing the input and calibrates its protocols on synthetic EEG with a planted ground truth (Section 4.1); Appendix G compares this design with the studies above element by element.

2.2 Foundations of Protocol Design: Neurophysiology and Controls

The eight protocols target three physiological properties of EEG: spatial topography and montage constraints (Pfurtscheller & Lopes da Silva, 1999; Ang et al., 2008; Stepnowsky et al., 2013; Davidson, 2004; Zheng & Lu, 2015), rhythm structure (Pfurtscheller & Lopes da Silva, 1999; Davidson, 2004; Zheng & Lu, 2015; Noachtar et al., 1999; Iber et al., 2007), and artifact and background dynamics (Fisher et al., 2014; Whitham et al., 2007; Lin et al., 2026). Physiology also names the fine structure in each: the μ and β rhythms over the central region in motor imagery (Pfurtscheller & Lopes da Silva, 1999; Ang et al., 2008), the microstructure of the α rhythm (Noachtar et al., 1999; Iber et al., 2007), and the seizure waveform independent of its amplitude (Noachtar et al., 1999). Every physical intervention has side effects (volume conduction, zero padding, added energy), so its control is designed in advance from the signal's physics, not from the outcome. Interpreting a reading only against its control has precedents: probe accuracy is checked against control tasks (Hewitt & Liang, 2019), attributions are tested by randomizing model and data (Adebayo et al., 2018), and observational studies screen confounding with negative controls (Lipsitch et al., 2010).

3 Methodology

3.1 Overview

The benchmark runs in two stages (Figure 1): Stage I locks one state per route under the same data, readout form, and checkpoint rule; Stage II trains no parameters and runs eight input-side protocols directly on that state. Outside the two stages, calibration and interface replacement form a closed loop. Calibration first re-runs the pipeline on synthetic EEG with planted dependencies to verify that readings recover the planted truth across spatial, temporal, and spectral domains (Section 4.1; Appendix E). Interface replacement then keeps the pretrained backbone frozen, replaces only the interface component (adapter or readout) that the profile points to, and re-runs the same protocol (Section 5.1; Appendix F). This distinguishes a defect of the interface from what remains unrecovered in the backbone.

3.2 Stage I: Locking One State per Route

We evaluate five released EEG-FMs, BIOT (Yang et al., 2023), CBraMod (Wang et al., 2025), CSBrain (Zhou et al., 2025), LaBraM (Jiang et al., 2024), and REVE (El Ouahidi et al., 2025), on four tasks, each with a well-defined neurophysiological mechanism: motor imagery (Motor), seizure detection (Seizure), sleep staging (Sleep), and emotion recognition (Emotion). The task metric Q is balanced accuracy for sleep staging (chance 0.2) and AUROC for the other three, one-vs-rest for the four-class tasks (chance 0.5). Signals follow standard preprocessing with subject-disjoint splits (Appendix A.1–A.2).

Each route's state comprises three components: a model-native adapter that handles channel mapping and the model's own amplitude transform, a frozen pretrained backbone, and a linear readout (Appendix A.3, A.4; Table A.3). The adapter and the readout together form the *interface* around the backbone. Only the readouts are trained, jointly fitted across the four tasks on the training split (Appendix A.5). The best checkpoint $\mathcal{C}_m$ is selected by mean validation balanced accuracy across tasks (Appendix A.6) and locked as the diagnostic state for Stage II.

3.3 Stage II: One Diagnosis End to End

All physical interventions in Stage II act on the $C \times T$ signal after montage alignment and prior to model-specific resampling and input representation (time-domain patches or STFT spectrograms; Appendix B.1). This section walks through protocol S1.1, *Region Criticality*, to illustrate how physical interventions combine with neurophysiological priors into a controlled causal intervention with paired controls.

S1.1 investigates whether model predictions rely on localized scalp organization (regional fine structure) rather than non-specific degradation from channel loss. The intervention masks one of five lobar regions at fractions $\rho \in \{0.25, 0.5, 0.75, 1\}$. To isolate performance drops from two non-target side effects, the loss of channel count and the numerical effect of zeroing itself, the design uses the unperturbed clean score as reference, pairs regional masking with a scattered control that randomly masks an equal number of channels outside the target region, and compares zeroing against temporal shuffling. Each response is summarized into two metrics:

$$\mathrm{RIS}_{r,\rho} = 1 - \frac{Q_{m,t}^{\mathrm{regional}(r,\rho)}}{Q_{m,t}^{\mathrm{clean}}}, \qquad \mathrm{SCS}_{r,\rho} = Q_{m,t}^{\mathrm{scattered}(r,\rho)} - Q_{m,t}^{\mathrm{regional}(r,\rho)},$$

the Regional Importance Score (RIS), the overall degradation relative to the clean score, and the Spatial Contrast Score (SCS), the loss beyond the scattered control, reported together with the clean lift above chance, $L_0 = Q(0) - Q_{\mathrm{ch}}$. A positive SCS is a necessary condition for attributing degradation to regional specificity.

All grids, controls, response metrics, and execution states are pre-specified (Appendix B.3). Test-set records assemble into self-contained Units $\boldsymbol{\psi}_{m,t,s}$ (Appendix B.6) and are interpreted in three ordered steps: (1) **Baseline observability:** check L_0 to rule out false robustness from floor effects; if clean performance is near chance, a flat response curve only reflects the floor effect, and the later steps are not interpreted; (2) **Target specificity:** check SCS to rule out non-specific channel loss; the drop must exceed that of the scattered control; and (3) **Implementation sensitivity:** compare zeroing against shuffling to rule out masking artifacts, confirming that readings reflect missing physiological content rather than operator-specific side effects.

Uncertainty is quantified at two levels: subject-level variability is captured via test-subject bootstrap confidence intervals; randomness across fitted states is quantified over $N = 3$ independent states, reporting mean and standard deviation to measure inter-state variation (Appendix B.7).

3.4 The Same Design Across Three Dimensions

The design of S1.1 applies to all protocols. Each protocol is formalized as a 6-tuple $S_s = (q_s, z_s, \mathcal{P}_s, \mathcal{K}_s, \mathcal{O}_s, g_s)$, specifying its question, target physiological dependency, parameterized intervention family, reference and controls, response specification, and assembly rules (Appendix B.3 fills all eight; a new protocol enters by filling the tuple).

The eight protocols test specific signal dependencies across three physical dimensions:

- D_1 **Spatial dimension:** S1.1 *Region Criticality* masks lobar regions to test local specificity; S1.2 *Hemispheric Asymmetry* swaps homologous left–right electrodes and preserves channel count; combined, they determine whether a model relies on regional fine structure or the coarse montage arrangement.
- D_2 **Temporal–spectral dimension:** S2.1 *Temporal Aperture* shortens the visible window and fills the rest with zeros (zero fill) or with repeats of the kept segment (repeat fill), testing whether short-window stationary statistics suffice; S2.3 *Frequency Sensitivity* sweeps low-pass cutoffs and band-stops to isolate the bands the decision rests on; S2.5 *Alpha Modulation* edits α-band power, bandwidth, and peak frequency to evaluate sensitivity to rhythm microstructure.
- D_3 **Artifact and background dimension:** S3.2 *Artifact Injection* evaluates artifact robustness; S3.3 *Background Crossfade* distinguishes reliance on target event waveforms from reliance on input amplitude; S3.4 *Rejection Entropy* measures representational dispersion when brain inputs are completely withdrawn.

All Units assemble into each model's **response profile**. For cross-architecture comparison, evidence is reported under standardized scale and coordinate conventions (Appendix B.5–B.6). Protocol specifications, controls, and applicability are detailed in Appendices B.3–B.4; Section 4 reports six of them, and S3.2 and S3.4 are in Appendix D.4–D.5.

4 Experimental Results and Diagnostic Evaluation

This section applies the two-stage pipeline of Section 3.

4.1 Calibration on Planted Ground Truth

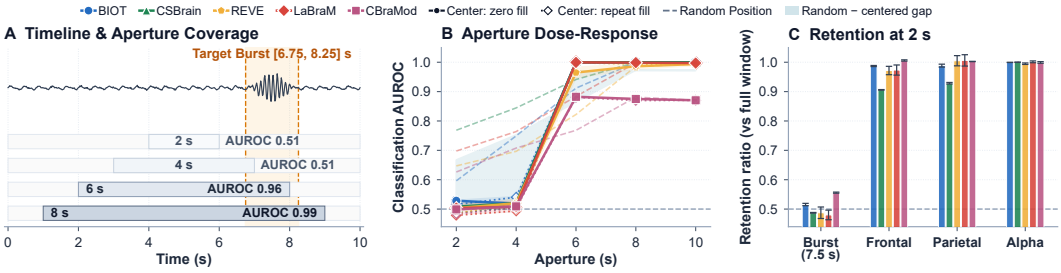


Figure 2: **Temporal-aperture calibration (S2.1) on planted bursts** (three seeds). (A) Aperture coverage. (B) AUROC vs. aperture length. (C) Retention at 2 s, burst vs. stationary tasks.

We first check the protocols against a known answer. Sinusoidal carriers planted in a known region, time window, or frequency band of synthetic EEG encode the class label, and three representative protocols are tested on two points: whether the reading detects the planted dependency, and whether it responds as expected under controlled fault injection (Appendix E). **Region criticality (S1.1).** Completely masking the planted region takes AUROC to chance (0.50–0.58), while the equal-channel scattered control changes it by at most 0.04. A temporal shuffle, which keeps channel energy, gives nearly the same Spatial Contrast Score as zeroing (SCS within 0.07). Under fault injection F1, a permuted channel map, the SCS turns negative (-0.09 to -0.18), so the reading follows the true position of each channel. **Temporal aperture (S2.1).** The reading depends on the signal energy the aperture captures, not on its length (Figure 2A–B). The burst is planted late, so a centered 4 s aperture captures 0.6% of its energy and a random 2 s aperture about 25%; the shorter aperture therefore scores higher (0.60–0.77 against 0.51–0.52). Zero fill and repeat fill differ by at most 0.033 on the burst but diverge on stationary signals, so a zero-fill loss on stationary content is read together with its paired repeat-fill control. **Frequency sensitivity (S2.3).** Performance drops only when the stop band covers a carrier: by 0.36–0.50 in the α band, which holds both carriers, and by 0.11–0.48 in the σ band, which holds one; the four carrier-free bands change by at most 0.027. Under fault injection F3, a mis-declared sampling rate, the drop moves to the nominal θ band (0.15–0.38), so the reading follows physical frequency, not the band label. All three protocols thus detect the planted dependency and respond to controlled faults as the physics predicts.

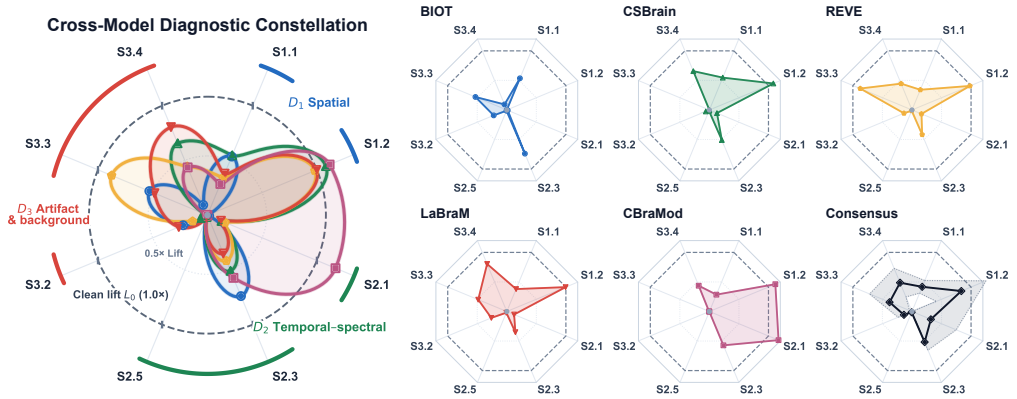


Figure 3: **Response profiles across eight diagnostic protocols.** Each axis shows the primary response used by that protocol; dashed lines show clean lift L_0. The axes use protocol-specific normalizations and should be read as a visual index rather than a common effect-size scale. Cross-model mean ± sd is shown for context; full scorecards are in Appendix C).

4.2 Response Profiles of Five Models

Response profiles show differences between models that a scalar task score hides (Figure 3). On Motor, CBraMod and CSBrain have nearly the same clean AUROC (0.750 and 0.748), yet complete frontal masking costs CSBrain almost twice as much as CBraMod (0.136 ± 0.026 against 0.071 ± 0.001). Each model also has its own most sensitive perturbation. BIOT responds most to low-pass filtering (a loss of 0.32 on Sleep, at $L_0 = 0.43$); CBraMod is highly sensitive to aperture truncation (a loss of 0.45 on Seizure, at $L_0 = 0.39$); and the models separate when non-neural noise replaces the recording in the background crossfade. Each vertex, however, records how strongly a model responds, not what the response rests on. Section 4.3 reads the responses against their controls, dimension by dimension, and finds one common preference: predictions rest mainly on coarse structure in the signal and respond weakly to neurophysiological fine structure.

4.3 Reliance on Coarse Structure, Weak Response to Fine Structure

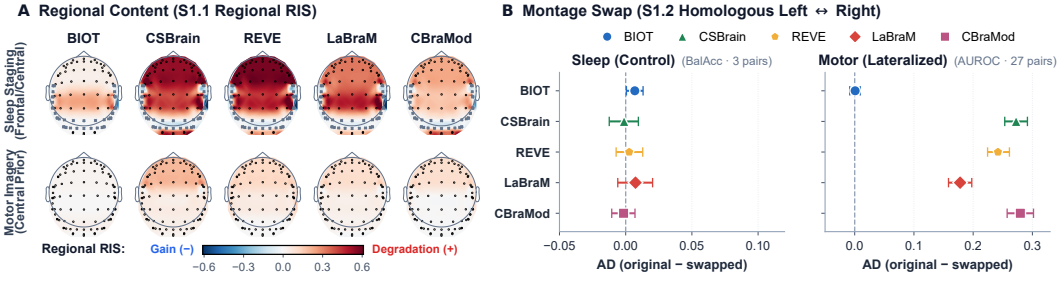


Figure 4: **Spatial reliance: regional content vs. montage arrangement.** (A) 10–10 projections of regional RIS (S1.1) on Sleep and Motor. (B) Hemispheric swap (S1.2): accuracy drop (AD) on Motor, and Sleep as the negative control.

Spatial dimension: regional content vs. montage arrangement. Predictions show weak region-specific reliance (Figure 4A). On Sleep, frontal masking costs CSBrain 0.274 ± 0.044 balanced-accuracy points, yet only SCS 0.030 ± 0.006 exceeds its equal-channel scattered control. On Motor, CBraMod and CSBrain lose most under frontal masking (0.071 ± 0.001 and 0.136 ± 0.026), with region-specific parts of only SCS 0.052 ± 0.001 and 0.085 ± 0.018. The central sensorimotor region (C3/Cz/C4, named by physiology) costs models with lift at most 0.034 ± 0.005 under complete masking, and exceeds its scattered control under neither zeroing nor shuffling. Across tasks, 206 of 1,080 regional maskings cost more than 0.05, and 32.0% of these are matched by their scattered

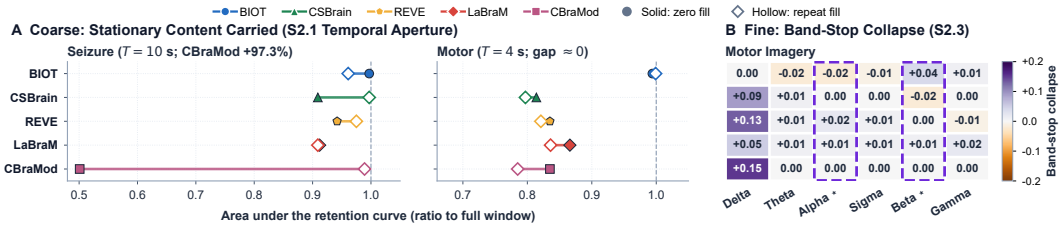


Figure 5: **Temporal–spectral reliance: stationary energy vs. physiological rhythms.** (A) Truncation retention on Seizure and Motor. (B) Motor band-stop loss (S2.3), design priors (μ, β) outlined.

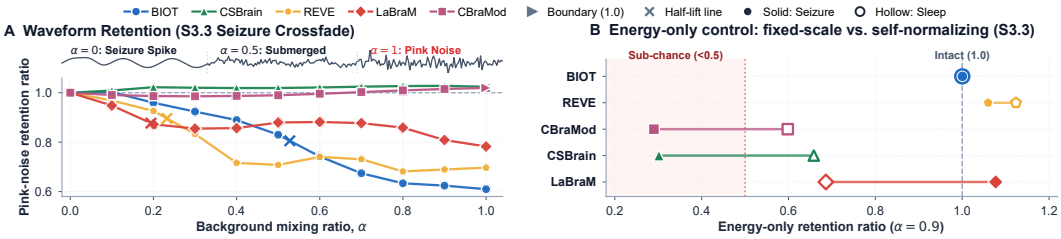


Figure 6: **Artifact and background reliance (S3.3).** (A) Seizure pink-noise retention curves. (B) Energy-only control at $\alpha = 0.9$, grouped by adapter scale.

control (Appendix C.2). Spatially, predictions thus respond weakly to the functional regions that physiology names.

Instead, spatial predictions rest on the montage arrangement (Figure 4B). On Motor, a task with a clear lateralization prior, the Hemispheric Asymmetry protocol (S1.2) keeps every channel and exchanges only the positions of homologous left and right electrodes. Every model with clean lift loses almost all of it (AUROC drop 0.18–0.28, against $L_0 \in [0.18, 0.26]$). On Sleep, the negative control without a lateralization prior, no model moves by more than 0.016 (Appendix D.1). The predictions therefore depend on the left–right arrangement, and only where the task is lateralized. This dependence is not the mirror that physiology predicts: CBraMod and CSBrain fall below chance (AUROC 0.471 and 0.480), yet reverse fewer than half of their left–right predictions ($44.3\% \pm 1.1\%$ and $39.8\% \pm 6.3\%$).

Temporal–spectral dimension: short-window stationary energy vs. physiological rhythms. In time, the predictions on Sleep and Seizure rest mainly on the stationary content of a short window. Repeat fill removes the step artifact that zero fill adds at the window boundary. It recovers 85–93% of the zero-fill loss on Sleep in every model (Appendix C.2) and $97.3\% \pm 0.1\%$ of CBraMod's loss on Seizure (0.453 ± 0.009 at $L_0 = 0.387$; Figure 5A), while the other four models respond weakly to truncation on Seizure (0.005–0.055). Motor, by contrast, needs the continuous time course: the four models with lift lose 0.080–0.136 under zero fill, and repeat fill does not recover the loss.

In frequency, the δ band (0.5–4 Hz), which carries most of the signal energy, drives the prediction more than any other band. A δ band-stop (S2.3) costs the four Motor models 6–15% of their reference and takes BIOT and REVE on Sleep to near chance (balanced accuracy $0.632 \rightarrow 0.209 \pm 0.002$ and $0.579 \rightarrow 0.230 \pm 0.050$, chance 0.20). The classical Motor rhythms, μ (8–13 Hz) and β (13–30 Hz), change the prediction by less than 2% when attenuated (Figure 5B). The band-stop can detect rhythm use where it exists: on Sleep, the β and σ band-stops cost BIOT 0.411 ± 0.009 and 0.188 ± 0.010. Editing the power, bandwidth, or peak frequency of the α band (S2.5) costs at most 0.019 on any task (Appendices D.2 and D.3). In the temporal–spectral dimension, the predictions thus rest on stationary energy in short windows and on the slow δ band, and respond weakly to the physiological rhythms and their microstructure.

Artifact and background dimension: channel amplitude vs. event waveform. Here some models use channel amplitude as a decision shortcut. In the background crossfade (S3.3), as pink noise is mixed into Seizure recordings, CBraMod and CSBrain never cross the half-lift line (Δ_{pink} 0.002 ± 0.001 and -0.016 ± 0.007, at $L_0 = 0.387$ and 0.370; Figure 6A). At full replacement ($\alpha = 1$),

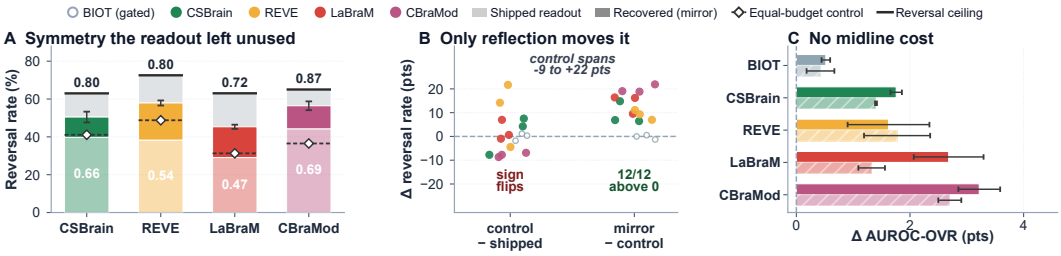


Figure 7: **Readout refit under hemispheric swap** (S1.2 Motor, three seeds). (A) Reversal rate vs. ceiling. (B) Mirror minus control per cell. (C) Clean AUROC over control; midline hatched.

the seizure waveform is gone and only noise matched to each channel's RMS amplitude remains, yet the two models still classify the input as seizure (AUROC 0.905 ± 0.003 and 0.894 ± 0.018). Conversely, the energy-only control, which only scales the recording by $1-\alpha$ and keeps the waveform intact, takes CBraMod and CSBrain below chance on Seizure at $\alpha = 0.9$ (AUROC 0.264 ± 0.010 and 0.323 ± 0.125) and removes 56–78% of the Sleep lift of the three fixed-scale models, CBraMod, CSBrain, and LaBraM (Figure 6B). Renormalizing each channel's amplitude, as in the S2.3 reference, also removes most of the Seizure lift of CBraMod and CSBrain (AUROC $0.887 \rightarrow 0.564$ and $0.870 \rightarrow 0.522$; Table D.1). BIOT and REVE move by at most 0.007 under this renormalization and lose at most 0.022 to the energy-only control, so their predictions depend more on waveform geometry than on amplitude. On Motor no model relies on amplitude: the loss under background replacement exceeds the loss under attenuation alone by 0.062–0.093 (Figure C.3). Where predictions track amplitude, they follow channel scale rather than the event waveform.

5 Discussion

5.1 At the Interface: Readout Refits and Deployment Guidelines

Under Hemispheric Asymmetry (S1.2), the models reverse fewer than half of their left–right predictions. Is the symmetry absent from the backbone's representation, or present and masked by the shipped readout? To answer it, we designed a readout refit (Figure 7; Appendix F.1).

The backbone is frozen and bit-identical across three readout states: the shipped readout; an equal-budget control, which continues training it for a fixed number of steps; and a mirror state, which continues it for the same number of steps on batches augmented with the left–right mirror, labels exchanged. Three observations follow. First, realization, the reversal rate over its achievable ceiling, rises from 0.47–0.69 for the shipped readout to 0.72–0.87 for the mirror state, while the ceiling itself barely moves (Figure 7A). Second, more training steps alone give no consistent gain, whereas mirror minus control is positive in all twelve model–seed cells (Figure 7B), so the gain is not due to the extra training. Third, on the unswapped test set the mirrored samples act as a regularizer: clean AUROC rises against the control in all twenty model–class means, the midline classes (both fists, both feet) included, by 0.016–0.030 on average for the four models with lift (Figure 7C).

A linear readout cannot build a feature that the backbone does not provide, so the symmetry was already in the frozen representation and the shipped readout had not used it. A low reading in a profile therefore need not mean that the representation lacks the property. Three guidelines follow: **Tell an unused capability from an absent one.** A low reading need not send a model back to pretraining; refit the readout first. What the refit recovers, the backbone held and the readout left unused; only what it does not recover is left to the backbone. With its readout refitted on injected ocular artifacts (Appendix F.2), LaBraM's ocular retention rises by 0.286, and by 0.272 at the doses withheld from the refit, so its backbone already separated that artifact from the signal; CSBrain's clean score and retention both fall, and the readout recovers nothing. Only the second case is left to next-generation pretraining (Section 5.2).

Write physiological priors into downstream training. Downstream training need not, by itself, use a structure the representation holds, as the equal-budget control shows. Adding a physiological prior

explicitly, for example through left–right mirror augmentation, puts that structure to use at the cost of training one linear readout.

Set preprocessing by the interface. On Sleep and Seizure, most of the loss under a shortened window comes from zero padding (Section 4.3), so fill a short window by repetition, not zeros. Where the adapter does not normalize each channel, keep the input at the scale used in training.

5.2 In the Backbone: Targets for Next-Generation Pretraining

A readout refit uses what the backbone provides but cannot add what it does not. The 13–28% of achievable reversals that the mirror state leaves unrealized, and CSBrain's failure under ocular artifacts, are not recovered at the readout and are left to the backbone. In each dimension, the profiles name one target for pretraining, the fine structure to which the models respond weakly (Section 4.3), and each target comes with a comparison under one pretraining recipe that could reject it (Table B.9): **Space:** from relying on the montage's left–right arrangement to using the regional content that physiology names; the test compares coordinate encoding with spatial pooling.

Temporal–spectral: from relying on the stationary content of a short window to using the microstructure of physiological rhythms; the test compares time-domain patches with STFT spectrograms.

Artifact and background: from relying on input amplitude to using the waveform independent of amplitude; the test compares fixed scaling with a gain-invariant objective.

5.3 Transfer and Limitations

The evidence here is on EEG, but the method of evaluation and improvement is not specific to one modality. For foundation models of other continuous physical signals, such as ECG, speech, or seismic recordings, it can transfer at two levels: **Causal interventions at the input, calibrated on synthetic ground truth.** The interventions act on the continuous physical signal, before each model's own input representation; paired controls separate out the side effects of each perturbation; and synthetic signals with a planted ground truth confirm that the readings recover it. Without these controls, a zero-padding loss would read as temporal dependence, and reliance on amplitude as robustness to background noise (Section 4.3).

Attribution by component, improvement at two levels. Freezing the backbone and replacing only the interface, here by refitting one linear readout, separates what the interface left unused from what it cannot recover. The first guides priors and preprocessing downstream; the second gives pretraining upstream a target with a test that could reject it. Without this step, the readout's failure under the swap would read as a backbone defect (Section 5.1). **Limitations.** We evaluate five released models in one frozen-linear state each and one dataset per task; Seizure and Emotion each contain three test subjects. Synthetic calibration covers three of the eight protocols, and several integrated response coordinates are reported as per-state estimates rather than fully propagated subject-level bootstrap quantities. The results should therefore be read as properties of the evaluated routes and datasets, rather than general properties of EEG foundation models. Broader validation requires additional cohorts, state families, baselines, and independently recorded artifacts.

6 Conclusion

Task scores tell us how well a model predicts, but not which properties of the signal support that prediction. Across five released EEG-FMs and four tasks, our controlled diagnostics reveal recurring sensitivities to left–right electrode swaps, short-window or low-frequency content, and channel scale that are largely hidden by clean scores. Paired controls help separate intended perturbations from major side effects, while synthetic experiments validate three representative protocols against known dependencies. Targeted readout refits further test which diagnosed behaviors can be changed without modifying the pretrained backbone. These results are specific to the evaluated routes and datasets, but they provide concrete, falsifiable hypotheses for downstream design and future pretraining. Dissecting the score therefore complements leaderboard evaluation with controlled tests of what an EEG-FM evaluation route uses and what can be changed after pretraining.

Reproducibility Statement

To support audit and reproduction, the release preserves subject-disjoint split manifests, pre-specified perturbation parameters, condition records, campaign allowlists, and completion receipts with explicit identities and status fields. Appendix H documents the observed software/runtime context, the Stage I training seeds `20260528`, `20260612`, and `20260718`, the separately scoped split/bootstrap and perturbation seeds, the canonical execution chain, and environment or checkpoint identities that were not captured. The execution codebase, protocol runners, condition-level logs, machine-readable Unit records, and available integrity receipts will be released with the publication artifact.

AI Use Statement

In compliance with the ICLR 2027 AI Policy for Authors, the authors disclose that large language models assisted with translating initial drafts into English, implementing and refactoring pipeline code, reformatting dataset records, critical review of methodological and interpretive claims, literature discovery, manuscript–artifact consistency checks, proofreading, and LaTeX formatting. They were not used to generate data, empirical observations, or scorecard values. The authors defined the research questions, hypotheses, mathematical formulations, perturbation designs, and diagnostic protocols; inspected the underlying records; adjudicated all AI suggestions; and reviewed and tested every retained claim and code change. The authors take full responsibility for the final content and the accuracy and integrity of the work.

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SUPPLEMENTARY APPENDIX

Each appendix answers one question from the main text.

A How is each model prepared and locked before diagnosis? Data, adapters, readouts, and checkpoint selection (Section 3.2).

B What does each protocol change, control, and measure? Intervention points, metric definitions, and one specification card per protocol (Sections 3.3–3.4).

C What stands behind the readings of Section 4? Task scorecards, the controls behind each reading, and one interface note per model.

D What do the five protocols that Section 4 does not cover in full show? Results for S1.2, S2.3, S2.5, S3.2, and S3.4.

E Do the protocols recover a known answer? Calibration on synthetic EEG with a planted ground truth (Section 4.1).

F Can a diagnosed dependence be repaired at the readout? Two readout refits against an equal-budget control (Section 5.1).

G How does this design relate to concurrent EEG-FM evaluations? A comparison table and its trade-offs (Section 2).

H What does reproduction need? Environment, seeds, and release records.

A STAGE I: DATA, ADAPTERS, READOUTS, AND THE LOCKED STATE

Main-text anchor: Section 3.2.

A.1 SOURCES AND SPLIT

Table A.1 lists the four sources, one per task; a task and its cohort therefore vary together, and this campaign cannot separate them. Every split is subject-disjoint,

$$\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{valid}} = \emptyset, \qquad \mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset, \qquad \mathcal{S}_{\text{valid}} \cap \mathcal{S}_{\text{test}} = \emptyset;$$

PhysioMI is split by subject index (1–69 / 70–88 / 89–109), and ISRUC, Siena, and SEED-IV draw their validation and test subjects once with seed 42. The release checks that subjects, recordings, sessions, and sample keys each lie in one partition, and that test rows are read only after the checkpoint is selected. These checks cannot see two things: the same person entering two sources under different identifiers, and a subject already seen during pretraining.

Table A.1: **Sources, tasks, and subject-disjoint partitions.** Storage is fixed at 256 Hz and μV; Stage I and Stage II share the same partitions.

					Train / validation / test	
Dataset[4]	**Task**[3]	K	**Montage**	$C \times T$ **(window)**	**Subjects**	**Windows**
PhysioMI	Motor Imagery	4	10–10	64×1024 (4 s)	69 / 19 / 21	6,210 / 1,734 / 1,893
ISRUC	Sleep Staging	5[1]	PSG	6×7680 (30 s)	80 / 9 / 9	71,955 / 8,350 / 8,103
Siena	Seizure Detection	2	10–20	29×2560 (10 s)	8 / 3 / 3	34,305 / 10,750 / 5,775
SEED-IV	Emotion Recognition	4	10–20[2]	60×2560 (10 s)	9 / 3 / 3	8,793 / 2,931 / 2,931

[1] ISRUC: 98 subjects; subjects 8 and 40 are excluded for unresolved required-channel metadata. [2] SEED-IV: 62 channels mapped to 60 standard 10–20 sites ($f_s = 200 \rightarrow 256$ Hz). [3] Later tables use the short identifiers `physiomi_mi`, `isruc_sleep`, `siena_seiz`, and `seed_iv_emo`; the codebase namespaces are `physiomi_motor_imagery`, `isruc_sleep_stage`, `siena_seizure`, and `seed_iv_emotion`. 236 subjects and 163,730 windows in total. [4] Sources: PhysioMI (Schalk et al., 2004; Goldberger et al., 2000), ISRUC (Khalighi et al., 2016), Siena (Detti et al., 2020), SEED-IV (Zheng et al., 2019).

A.2 Storage Convention

Every recording is processed once and stored in one form; each adapter converts from that form at run time:

$$x_d^{\text{raw}} \xrightarrow{\text{channel select / alias}} x_d^{(1)} \xrightarrow{\text{band-pass + notch + resample}} x_d^{(2)} \xrightarrow{\text{window segmentation}} x_d^{\text{storage}} \quad (f_s = 256\,\text{Hz},\ \mu\text{V}).$$

Every recording is resampled with anti-aliasing (MNE) to 256 Hz and stored in μV without normalization; fixed scaling and per-window standardization belong to the first adapter layer (Table A.2). Each source is band-passed to its own pass band—PhysioMI 0.1–100 Hz, ISRUC 0.3–35 Hz, Siena 0.1–95 Hz, SEED-IV 0.1–75 Hz—with the upper cutoff applied only where the original sampling rate allows it. Line noise is notched at 60 Hz for PhysioMI and 50 Hz for the other three, with harmonics below the original Nyquist frequency. Channels keep 10–10 / 10–20 names; a channel the montage lacks stays absent and is never zero-padded in storage.

A.3 Model-Native Adapters

The five models are CSBrain, LaBraM, BIOT, CBraMod, and REVE (Zhou et al., 2025; Jiang et al., 2024; Yang et al., 2023; Wang et al., 2025; El Ouahidi et al., 2025), each with a public pretrained checkpoint and a deterministic adapter. An adapter has two layers. The first receives the stored 256 Hz tensor, selects the channels the model supports, and applies the model's own amplitude transform; the second resamples to 200 Hz, builds the model's montage arrangement, and forms the input representation its backbone consumes (Table A.2). Stage II reuses both layers and adds only the intervention placements and normalization settings of Appendix B.1.

Table A.2: **Adapter layers per model ($M = 5$).** All five routes resample linearly to 200 Hz in the second layer.

Model	Amplitude	Channels	Input representation and positions
CSBrain	Fixed $\times 0.01$	Supported subset; five-region grouping	1 s non-overlapping patches; region-aware spatial PE
LaBraM	Fixed $\times 0.01$	Standard 10–20 channels, indexed	1 s non-overlapping patches; 1D spatial + temporal PE[2]
BIOT	Identity[1]	Fixed 18-channel bipolar montage (missing pairs zero-padded)	1 s STFT (0.5 s hop); 95th-percentile scaling; mean-pooled output
CBraMod	Fixed $\times 0.01$	Fixed 10–10 sequence	1 s non-overlapping patches; adaptive convolutional PE
REVE	Per-window z-score	Retained in bank order	1 s patches (0.1 s overlap); 4D Fourier coordinates

[1] BIOT divides each derivation by its own 95th percentile before the STFT. [2] LaBraM truncates or interpolates its released temporal-position table to the window length.

A.4 Readout

One affine layer reads the output features the backbone exposes,

$$h_{\phi_{m,t}}(\mathbf{Z}) = W_{m,t}\,\text{vec}(\mathcal{L}_m(\mathbf{Z})) + b_{m,t}, \qquad W_{m,t} \in \mathbb{R}^{K_t \times (P_{m,t} C_{m,t} d_{h,m})},\ b_{m,t} \in \mathbb{R}^{K_t}, \quad (1)$$

where $\mathcal{L}_m$ arranges the output as batch–time–channel–feature, K_t is the number of classes, and P, C, d_h are the route's temporal patch count, channel count, and feature width; BIOT exposes one mean-pooled vector and enters as $(P, C, d_h) = (1, 1, 256)$. The head adds no pooling, hidden layer, or nonlinearity. Table A.3 gives the dimensions and parameter counts $K(PCd_h + 1)$, which run from 514 (BIOT, Seizure) to 1.35×10^6 (REVE, Emotion) and are reported rather than equalized. Within one Unit the head is fixed, so a difference between conditions is the response of a single state, not a change in head capacity. A difference between models is a difference between whole routes; this campaign does not separate the contributions of backbone, pretraining, adapter, and readout.

Table A.3: **Readout dimensions and trainable parameter counts per route.** P is the temporal patch count, C the retained channel count, d_h the feature width, and K the number of classes.

Task (K, Window, Channels)	BIOT	CBraMod	CSBrain	LaBraM	REVE
Sleep Staging ($K = 5, 30\,\text{s}, 6\,\text{ch}$)					
Route Shape (P, C, d_h)	$(1, 1, 256)$	$(30, 6, 200)$	$(30, 6, 200)$	$(30, 6, 200)$	$(33, 6, 512)$
Flattened Dimension (PCd_h)	256	36,000	36,000	36,000	101,376
Trainable Parameters	**1,285**	**180,005**	**180,005**	**180,005**	**506,885**
Motor Imagery ($K = 4, 4\,\text{s}, 64\,\text{ch}$)					
Route Shape (P, C, d_h)	$(1, 1, 256)$	$(4, 64, 200)$	$(4, 64, 200)$	$(4, 64, 200)$	$(4, 64, 512)$
Flattened Dimension (PCd_h)	256	51,200	51,200	51,200	131,072
Trainable Parameters	**1,028**	**204,804**	**204,804**	**204,804**	**524,292**
Emotion Recog. ($K = 4, 10\,\text{s}, 60\,\text{ch}$)					
Route Shape (P, C, d_h)	$(1, 1, 256)$	$(10, 60, 200)$	$(10, 60, 200)$	$(10, 60, 200)$	$(11, 60, 512)$
Flattened Dimension (PCd_h)	256	120,000	120,000	120,000	337,920
Trainable Parameters	**1,028**	**480,004**	**480,004**	**480,004**	**1,351,684**
Seizure Detection ($K = 2, 10\,\text{s}, 29\,\text{ch}$)					
Route Shape (P, C, d_h)	$(1, 1, 256)$	$(10, 29, 200)$	$(10, 29, 200)$	$(10, 29, 200)$	$(11, 29, 512)$
Flattened Dimension (PCd_h)	256	58,000	58,000	58,000	163,328
Trainable Parameters	**514**	**116,002**	**116,002**	**116,002**	**326,658**

A.5 Readout Training

The four readouts of a model are fitted together with the backbone in evaluation mode,

$$\phi_m^* = \arg \min_{\{\phi_{m,t}\}_{t=1}^{T}} \mathbb{E}_{t \sim \mathcal{E}_{\text{phys}}} \mathbb{E}_{(x,y) \sim \mathcal{D}_t^{\text{train}}} \left[\ell_t \big(h_{\phi_{m,t}} (f_{\theta_m^{\text{pre}}} (A_m(x))), y \big) \right], \tag{2}$$

where A_m is the adapter and ℓ_t the cross-entropy of task t. Each mini-batch holds windows from one source, and its loss updates only that source's head. The schedule $\mathcal{E}_{\text{phys}}$ gives every source the same per-epoch budget of tensor bytes and rotates which of a large source's batches fill it, so an epoch balances signal volume rather than window, subject, or label counts; the settings are in Table A.4.

Table A.4: **Stage I readout training.** All backbones are frozen in evaluation mode ($\nabla_\theta \mathcal{L} \equiv 0$).

Setting	Value	Note
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)	Decoupled weight decay 1×10^{-2}
Learning rate	Peak 1×10^{-4}; floor 1×10^{-5}	2-epoch linear warmup from the floor, then cosine decay to it
Gradient clipping	Global L_2 norm ≤ 1.0	
Batch size	32 windows	Remainder batches kept
Budget	25 epochs; no early stopping	Checkpoint chosen over all 25 epochs (Appendix A.6)
Loss	Cross-entropy, label smoothing $\alpha = 0.1$	Unweighted
Precision	bfloat16 autocast with gradient scaling	
Source schedule ($\mathcal{E}_{\text{phys}}$)	Equal per-source tensor-byte budget; rotating batch offset	Source-homogeneous batches
Label floor	Siena seizure windows $\geq 15\%$ of that source's epoch slots	The only floor applied; filled by within-source replacement

A.6 Checkpoint Selection and Locking

After each epoch e the validation partition gives one score per model,

$$\text{score}(e) = \frac{1}{|\mathcal{N}_e|} \sum_{t \in \mathcal{N}_e} \text{BalancedAcc}_{m,t}^{\text{val}}(e), \tag{3}$$

over the tasks $\mathcal{N}_e$ with a finite validation balanced accuracy; a checkpoint replaces the incumbent only on a strict improvement, so a later tie keeps the earlier epoch. The selected checkpoint—backbone and four heads—is serialized with the run configuration, from which the adapter is rebuilt. Before Stage II acquires any response, three conditions are enforced: the checkpoint, the Stage I configuration, the selection record, and the evaluation manifest are bound by SHA-256 digests; the selection bank is `natural_valid`; and no test row has been read. Stage II then reads the locked state with evaluation-only forward passes on `natural_test` and changes no parameter.

A.7 State Family

Stage I declares a state as an adapter, a training regime, and a readout family (Table A.5). This campaign runs one state per route, the frozen backbone under an affine readout; a fine-tuned backbone or a nonlinear head enters as another declared state and leaves the protocols untouched.

Table A.5: **Declared state family.**

State	Trained parts	In this campaign
Frozen backbone, affine readout ($\mathcal{C}_m$)	Readout heads	Run for all five models and all eight protocols
Fine-tuned backbone	Backbone and heads	Declared; not run
Nonlinear readout (MLP or attention head)	Readout heads	Declared; not run

B Stage II: Protocols

Main-text anchors: Sections 3.3–3.4.

The eight protocols are S1.1 and S1.2 (D_1), S2.1, S2.3, and S2.5 (D_2), and S3.2, S3.3, and S3.4 (D_3); the identifiers are repository indices and are not contiguous.

B.1 Where the Intervention Acts

Every intervention changes the $C \times T$ signal after channel alignment and before the model's own resampling and input representation; backbone and readout stay frozen. Table B.1 places each protocol relative to the adapter's amplitude transform and resampling.

Table B.1: **Intervention point per protocol**, relative to the adapter's first-layer amplitude transform and second-layer resampling.

Protocol	What is changed	Rel. to amplitude transform	Rel. to resampling
S1.1 Region Criticality	Supported-channel tensor	After	Before
S1.2 Hemispheric Asymmetry	Homologous left–right pairs	After	Before
S2.1 Temporal Aperture	Full exchange window	Before	Before
S2.3 Frequency Sensitivity	Full-band waveform	Before[1]	Before
S2.5 Alpha Modulation	Fourier alpha spectrum	Before	Before
S3.2 Artifact Injection	Additive EMG / EOG waveforms	Before[2]	Before
S3.3 Background Crossfade	Background mixture	Before[1]	Before
S3.4 Rejection Entropy	Whole input	After	Before

[1] S2.3 replaces adapter z-scoring by an explicit per-channel RMS renormalization applied to every filtered condition and to the full-spectrum reference; S3.3 switches z-scoring off so that the energy-only control is not renormalized away. Fixed scalar scaling stays as each model declares it. [2] S3.2 mixes the artifact in μV before the model's scaling.

B.2 Notation and Pipeline

A protocol fixes its six slots before any response is acquired. Its conditions are then evaluated on one locked state, read in the order of Table B.3, and assembled into one Unit per model–task pair (Appendix B.6). The pipeline runs

$$\mathcal{D}^{\text{train}} \xrightarrow{\text{Prepare}} \mathcal{C}_m \xrightarrow{\{S_s\}_{s=1}^{S}} \mathbf{\Phi}_m, \qquad \mathbf{\Phi}_m \coloneqq \{\boldsymbol{\psi}_{m,t,s}\}_{t,s}\,, \tag{4}$$

where $\mathcal{C}_m = (A_m, f_{\theta_m^{\text{pre}}}, \{h_{\phi^*_{m,t}}\})$ is the locked state of model m and $\mathbf{\Phi}_m$ its response profile. A protocol is the tuple

$$S_s = (q_s, z_s, \mathcal{P}_s, \mathcal{K}_s, \mathcal{O}_s, g_s)\,, \tag{5}$$

its question, target dependency, intervention family with grid, reference and controls, response specification, and assembly rule. Calligraphic symbols are design-time specifications; upright bold symbols are acquired records (Table B.2).

Table B.2: **Notation.**

Symbol	Meaning
m, t, s, a, λ	Model, task, protocol, condition, and intervention coordinate.
$\mathcal{D}^{\text{train}}, \mathcal{D}^{\text{val}}, \mathcal{D}^{\text{test}}$	Subject-disjoint partitions (Appendix A.1).
A_m	Model-native adapter (Appendix A.3).
$f_{\theta_m^{\text{pre}}}$	Pretrained backbone, frozen in evaluation mode.
$h_{\phi^*_{m,t}}$	Locked affine readout of route (m, t).
$\mathcal{C}_m$	Locked state $(A_m, f_{\theta_m^{\text{pre}}}, \{h_{\phi^*_{m,t}}\})$.
$S_s = (q, z, \mathcal{P}, \mathcal{K}, \mathcal{O}, g)$	Protocol design tuple.
$\mathcal{A}_s, \Lambda_s$	Condition set and intervention grid of protocol s.
$\mathbf{O}_{m,t}^{s,a,\lambda}$	Condition record at condition a and coordinate λ.
$\boldsymbol{\psi}_{m,t,s}, \mathbf{\Phi}_m$	Unit, and the response profile of model m.
Q, Q_{ch}, L_0	Task metric, its chance level, and the clean lift $Q(0) - Q_{\text{ch}}$.

B.3 The Eight Protocols in Six Slots

Table B.3 names the four roles under which a Unit is read and the release field that records each. Table B.4 gives the question, the target, the reference, and the controls of each protocol, and the cards of Sections B.8–B.15 fix the grid and the responses. A protocol specifies a control ($\mathcal{K}_s$) only where the intervention has a side effect that a matched condition can isolate; where no condition can isolate it, the limit is stated as applicability instead. All six slots, the state they run on, and the code identity are bound before the first response is acquired.

Table B.3: **The four roles under which a Unit is read, and the release field that records each.**

Role	Release field	What supplies it
Baseline observability	`baseline_capacity`	$Q(0)$, Q_{ch}, and L_0 of the route
Target specificity	`non_target_information_loss`	A matched non-target change of the same size (S1.1 scattered masking; S3.3 energy-only attenuation; S2.3 band-stops outside the target band)
Implementation sensitivity	`intervention_artifact`	Two realizations of one intervention (S1.1 zero / shuffle; S2.1 zero / repeat fill, centered / random placement; S3.3 white / phase-scrambled backgrounds)
Scope and applicability	`measurement_scope`	Where the measurement is defined (montage coverage, complete pairs, target channels, the absence of a clean-EEG reference in S3.4)

Table B.4: **The eight protocols: question, target, reference, and pre-specified controls.** Grids and response specifications are fixed once in the protocol cards (Sections B.8–B.15). Every protocol assembles one Unit per model–task pair; S3.4 pools the four montages into one Unit per model.

Protocol	q: question	z: target	Ref.	$\mathcal{K}$: controls
S1.1 Region Criticality	Does the route depend on localized scalp organization beyond the loss of an equal number of channels?	Lobar spatial organization	Clean score	Equal-channel scattered masking; shuffle against zero
S1.2 Hemispheric Asymmetry	Does the route depend on left–right spatial order?	Bilateral asymmetry	Original montage	None; the complete-pair check is applicability
S2.1 Temporal Aperture	How much temporal context does the route need?	Temporal context sufficiency	Full window	Repeat against zero fill; random against centered placement
S2.3 Frequency Sensitivity	Which bands carry the decision?	Band sufficiency and necessity	Full spectrum	Band-stops outside the target band measure the filter's own side effect
S2.5 Alpha Modulation	Is the route sensitive to the geometry of the alpha band?	Alpha power, bandwidth, peak frequency	Identity coordinate	None
S3.2 Artifact Injection	How does the decision degrade with artifact dose?	Artifact dose	Clean score	None
S3.3 Background Crossfade	Can the route extract events as the background rises?	Signal–background separation	$\alpha = 0$	White noise; phase-scrambled pink; energy-only attenuation
S3.4 Rejection Entropy	How dispersed is the representation for non-brain input?	Representation dispersion	`original_eeg` label	None; no clean-EEG reference exists

B.4 Applicability

Table B.5 states, per task, where each protocol's physiological reading is defined; it bounds the reading and is fixed before acquisition.

Table B.5: **Task $\times$ protocol applicability.**

Task	S1.1	S1.2	S2.1	S2.3	S2.5	S3.2	S3.3	S3.4
Motor	Central	Positive	✓	μ/β (8–30 Hz)	✓	✓	✓	✓
Sleep	Frontal/central	Negative control	✓	δ, σ	Conditional	EOG only[1]	✓	✓
Seizure	Temporal	Uncertain	✓	δ, γ	Conditional	✓	✓	✓
Emotion	Frontal/temporal	Cond. (FAA)	✓	β, γ	Conditional	✓	✓	✓

✓: applicable without condition; a named region or band is the pre-specified target. [1] The ISRUC montage has no peripheral temporal channels, so EMG injection is `not_applicable`. S3.4 runs under each task's montage and assembles one pooled Unit per model.

B.5 Responses and Derived Coordinates

Q is AUROC for Seizure Detection, one-versus-rest AUROC for Motor Imagery and Emotion Recognition, and balanced accuracy for Sleep Staging, with $Q_{\mathrm{ch}} = 0.5$ for all AUROC metrics and 0.2 for balanced accuracy; S3.4 reads representation geometry and has no task chance level. Table B.6 defines every coordinate. Curve protocols are summarized by the signed mean loss $\bar{\Delta}_s = Q(0)[1 - \mathrm{AURC}]$ on the task metric's own scale and read beside L_0. For AURC, $\tilde{\lambda} = [T_s(\lambda) - T_s(\lambda_{\min})]/[T_s(\lambda_{\max}) - T_s(\lambda_{\min})]$ with $T_s(\lambda) = \log \lambda$ on the S2.3 cutoff axis and $T_s(\lambda) = \lambda$ elsewhere. Clipping applies only to the coordinates labeled clipped; a coordinate whose denominator vanishes ($Q(0) = 0$ or $L_0 = 0$) is left undefined, and its condition record is kept.

Table B.6: **Response metrics and derived coordinates.** $Q(0)$ is the clean score, Q_{ch} the chance level, and $L_0 = Q(0) - Q_{\mathrm{ch}}$ the clean lift; A_a is the raw AURC of background a (pink p, white w, phase-scrambled ph, energy-only e).

Metric	Definition	Meaning	Protocol
Retention and curve summaries			
Raw retention R_{raw}	$Q(\lambda)/Q(0)$	Score relative to clean	S2.1, S2.3, S2.5, S3.3
Adjusted retention R_{adj}	$[Q(\lambda) - Q_{\mathrm{ch}}]/L_0$	Chance-corrected retention	S2.1
Clipped AURC	$\int_0^1 \mathrm{clip}(R(\tilde{\lambda}), 0, 1)\, d\tilde{\lambda}$	Lower = stronger response; gains clipped	S2.1, S2.3
Raw AURC	$\int_0^1 R(\tilde{\lambda})\, d\tilde{\lambda}$	Unclipped; values > 1 are gains	S2.5, S3.3
Signed mean loss $\bar{\Delta}_s$	$Q(0)\,[1 - \mathrm{AURC}]$	Metric points; negative = average gain	S2.1, S2.3, S2.5, S3.3
Spatial (D_1)			
Regional importance RIS	$1 - Q^{\mathrm{regional}}(r, \rho)/Q^{\mathrm{clean}}$	Higher = larger regional loss	S1.1
Regional endpoint loss Δ_r	$Q^{\mathrm{clean}} - Q^{\mathrm{regional}}(r, 1)$	Loss at full regional masking	S1.1
Zero–shuffle gap ZSG	$\lvert \mathrm{RIS}^{\mathrm{zero}}_{r,\rho} - \mathrm{RIS}^{\mathrm{shuffle}}_{r,\rho} \rvert$	Zeroing against temporal shuffling	S1.1
Spatial contrast SCS	$Q^{\mathrm{scattered}}(r, \rho) - Q^{\mathrm{regional}}(r, \rho)$	Regional against equal-channel scattered masking	S1.1
Area under RIS ARIC	$\int_0^1 \mathrm{RIS}_{r,\rho}\, d\rho$	Area under the RIS curve	S1.1
Prediction reversal PRR	Strict left↔right reversals / true left/right trials	Reversal rate under the swap	S1.2
Accuracy drop AD	$Q^{\mathrm{original}} - Q^{\mathrm{swapped}}$	Loss under the homologous swap	S1.2
Temporal and spectral (D_2)			
Aperture retention t_{90}	$\min\{t : R(t) \geq 0.9\}$, in seconds	Shortest aperture with 90% retention	S2.1
Padding gap Δ_{pad}	$\mathrm{AURC}_{\mathrm{repeat}} - \mathrm{AURC}_{\mathrm{zero}}$	Fill sensitivity	S2.1
Position gap APG	$\mathrm{AURC}_{\mathrm{random}} - \mathrm{AURC}_{\mathrm{center}}$	Positive favors random placement	S2.1
Band-gain profile $R(f)$	$Q(\mathrm{lp}_f)/Q(\mathrm{full})$	Cumulative low-pass sufficiency	S2.3
Band collapse b	$1 - R(\mathrm{bandstop}_b)$	Drop when band b is attenuated	S2.3
High-band gap	$1 - R(45)$	Reliance above 45 Hz	S2.3
Axis AURC	$\mathrm{AURC}_{\mathrm{power}}$, $\mathrm{AURC}_{\mathrm{bw}}$, $\mathrm{AURC}_{\mathrm{peak}}$	Unclipped response along each alpha axis	S2.5
Modulation asymmetry MRA	$\max_a \max_d \lvert R_a(r_a + d) - R_a(r_a - d) \rvert$	Asymmetry around the identity coordinate	S2.5
Artifact and background (D_3)			
Artifact gap Δ_a	$Q_{\mathrm{clean}} - Q_a(\mathrm{SNR}_{\mathrm{ref}})$	Drop at the reference ratio 2	S3.2
Cross-artifact gap CAD	$\lvert \Delta_{\mathrm{EMG}} - \Delta_{\mathrm{EOG}} \rvert$	Type disparity; the two types use different channel counts	S3.2
Artifact dose AUC $\mathrm{AUC}_a^{\log}$	$\frac{1}{\log_{10} 20} \int_{0.5}^{10} \Delta_a(s)\, d\log_{10} s$, trapezoidal	Dose response on the log axis; the scorecard quantity	S3.2
Event extraction threshold EET	α where $Q(\alpha) = Q_{\mathrm{ch}} + 0.5 L_0$, interpolated	First descending half-lift crossing; 0 or 1 if the scan starts below or never crosses	S3.3
Spectrum gap	$A_p - A_w$	Pink against white background	S3.3
Phase gap	$A_p - A_{\mathrm{ph}}$	Pink against phase-scrambled background	S3.3
Energy-confound gap	$A_e - A_p$	Attenuation alone against pink background	S3.3
Normalized entropy H_{norm}	$(-\sum_i p_i \log p_i)/\log d_h$	Covariance-spectrum dispersion over feature dimension d_h	S3.4

B.6 CONDITION RECORDS AND UNITS

With $x^{\mathrm{in}}_{m,t}$ the stored input, $\mathcal{P}^{(a)}_{s,\lambda}$ the transformation of condition a at coordinate λ, and $A^{s,\mathrm{post}}_m$ the adapter stages after the intervention point,

$$\widetilde{x}^{s,a,\lambda}_{m,t} = \mathcal{P}^{(a)}_{s,\lambda}(x^{\mathrm{in}}_{m,t}), \quad \mathbf{Z}^{s,a,\lambda}_{m,t} = f_{\theta^{\mathrm{pre}}_m}\Big(A^{s,\mathrm{post}}_m(\widetilde{x}^{s,a,\lambda}_{m,t})\Big), \quad \widehat{\mathbf{y}}^{s,a,\lambda}_{m,t} = h_{\phi^{\star}_{m,t}}(\mathbf{Z}^{s,a,\lambda}_{m,t}), \tag{6}$$

and the response functional μ_s turns predictions, optional representations, and labels into a condition record, from which the Unit is assembled:

$$Y^{a}_{m,t,s}(\lambda) = \mu_s\Big(\widehat{\mathbf{y}}^{s,a,\lambda}_{m,t}, \mathbf{Z}^{s,a,\lambda}_{m,t}; \mathbf{y}_t\Big), \qquad \boldsymbol{\psi}_{m,t,s} = \mathrm{AssembleUnit}\Big(\{\mathbf{O}^{s,a,\lambda}_{m,t}\}_{a,\lambda}; S_s\Big). \tag{7}$$

A condition record stores the coordinate, the measured response and its derived coordinates, a measurement status, audit flags, and provenance; every expected coordinate has exactly one status, `measured`, `not_applicable`, `missing`, or `failed`, and a run is complete only when no coordinate is missing or failed and its code identity is clean. The Unit adds the reference and its lift, the per-condition curves, the pre-specified contrasts, uncertainty status, and links back to its records. The view below shows these pieces for one Unit (CBraMod, Motor Imagery, S1.1). The released files group them under the nested blocks `identity`, `measurement`, `measurement_context`, `uncertainty`, `audit`, `analysis_trace`, and `provenance`; the scorecards of Appendix C extract the same endpoint from each seed's Unit and then summarize across seeds.


```
{
  "unit_id": "unit_CBraMod_physiomi_mi_S1.1_v1",
  "provenance": {
    "model": "CBraMod",
    "task": "Motor_Imagery",
    "dataset": "PhysioNet_MI",
    "protocol": "S1.1_Region_Criticality"
  },
  "baseline_observability": {
    "clean_metric": "AUROC_OvR",
    "clean_score": 0.750,
    "chance_level": 0.500,
    "baseline_lift_L0": 0.250
  },
  "condition_records": {
    "target": {
      "region": "frontal",
      "channel_count": 24,
      "endpoint_loss": 0.071,
      "measurement_status": "measured"
    },
    "scattered_control": {
      "type": "equal_channel_scattered",
      "channel_count": 24,
      "endpoint_loss": 0.019,
      "measurement_status": "measured"
    }
  },
  "assembled_contrasts": {
    "SCS_spatial_specificity": 0.052
  },
  "uncertainty_and_audit": {
    "target_seed_std_N3": 0.001,
    "perturbation_audit": {
      "atlas_id": "eeg_1010_lobar_prefix_v1",
      "selection_strategy": "seeded_nested_random_v1",
      "effective_ratio": 1.0,
      "eligible_count": 24,
      "status": "applied"
    }
  }
}
```

B.7 UNCERTAINTY

Condition-level intervals are percentile intervals from test-subject bootstrap resampling ($B = 1000$, 95%, seed 42). Scorecards report mean ± sample standard deviation over three independently fitted states (seeds `20260528`, `20260612`, `20260718`); this spread measures variation between fitted states under fixed splits and fixed perturbation realizations. Integrated coordinates (RIS, SCS, AURC, EET, Δ_{pad}) are per-state point estimates, since aligned subject resamples were not propagated through the surfaces; reported p-values are descriptive and uncorrected. Table B.7 gives the clean-score intervals of every route.

Table B.7: **Clean-score subject-bootstrap intervals per route** ($B = 1000$, 95% percentile). †: three test subjects, so the resampling support is discrete.

Task	Model	Metric	Point Estimate	95% CI Low	95% CI High	Test Subj. (N)
Motor Imagery	BIOT	AUROC-OvR	0.5052	0.4879	0.5209	21
(`physiomi_mi`, 1893 windows)	CBraMod	AUROC-OvR	0.7506	0.7197	0.7801	21
	CSBrain	AUROC-OvR	0.7528	0.7193	0.7901	21
	LaBraM	AUROC-OvR	0.6895	0.6621	0.7178	21
	REVE	AUROC-OvR	0.7635	0.7291	0.7950	21
Sleep Staging	BIOT	Balanced Acc.	0.6359	0.5817	0.6831	9
(`isruc_sleep`, 8103 windows)	CBraMod	Balanced Acc.	0.4126	0.3234	0.5026	9
	CSBrain	Balanced Acc.	0.5547	0.5043	0.5954	9
	LaBraM	Balanced Acc.	0.5069	0.4785	0.5357	9
	REVE	Balanced Acc.	0.5583	0.5149	0.6010	9
Emotion Recognition†	BIOT	AUROC-OvR	0.4868	0.4054	0.5252	3†
(`seed_iv_emo`, 2931 windows)	CBraMod	AUROC-OvR	0.5306	0.5201	0.5455	3†
	CSBrain	AUROC-OvR	0.5559	0.5496	0.5654	3†
	LaBraM	AUROC-OvR	0.5616	0.5514	0.5752	3†
	REVE	AUROC-OvR	0.5568	0.5544	0.5965	3†
Seizure Detection†	BIOT	AUROC	0.8186	0.6534	0.9596	3†
(`siena_seiz`, 5775 windows)	CBraMod	AUROC	0.8840	0.8720	0.9762	3†
	CSBrain	AUROC	0.8561	0.8304	0.9744	3†
	LaBraM	AUROC	0.6630	0.4179	0.7914	3†
	REVE	AUROC	0.8069	0.7224	0.9245	3†

B.8 S1.1 REGION CRITICALITY (D_1)

- **Grid:** the atlas `eeg_1010_lobar_prefix_v1` partitions the montage into frontal, central, temporal, parietal, and occipital groups (Table B.8); 5 regions × $\rho \in \{0.25, 0.50, 0.75, 1.00\}$ × $\{\text{zero}, \text{shuffle}\}$.
- **Reference and controls:** clean score; one seeded scattered subset per condition (subset variance not propagated); zeroing against shuffling.
- **Quantities:** $\mathrm{RIS}_{r,\rho} = 1 - Q^{\mathrm{regional}}(r,\rho)/Q^{\mathrm{clean}}$; $\mathrm{SCS}_{r,\rho} = Q^{\mathrm{scattered}}(r,\rho) - Q^{\mathrm{regional}}(r,\rho)$.
- **Applicability:** a region without montage channels is `not_applicable`; the reading is an operational electrode partition, not a source location.
- **Anchors:** Section 4.3; Figures 4 and C.1; Table C.1.

Table B.8: **Channels per scalp region in S1.1.** Masking acts on the supported-channel tensor (Appendix B.1); n_r is shared across models for a given task.

Task (Montage)	Frontal	Central	Temporal	Parietal	Occipital	Total
Motor Imagery (PhysioMI, 64 ch)	24	14	8	14	4	64
Sleep Staging (ISRUC, 6 ch)	2	2	—	—	2	6
Emotion Recog. (SEED-IV, 60 ch)	21	14	6	16	3	60
Seizure Det. (Siena, 29 ch)	13	7	2	5	2	29

B.9 S1.2 Hemispheric Asymmetry (D_1)

- **Grid:** the 90-channel registry defines 40 homologous left–right pairs and 10 midline channels; each montage swaps only the complete pairs it contains and leaves midline and unpaired channels in place (mean pair counts 3 / 27 / 26 / 13 for Sleep / Motor / Emotion / Seizure).
- **Reference and controls:** original montage; Sleep Staging is the pre-specified negative-control task.
- **Quantities:** PRR with a Wilson interval; $\mathrm{AD} = Q^{\mathrm{original}} - Q^{\mathrm{swapped}}$ with a class-stratified paired bootstrap and a paired one-sided randomization test.
- **Applicability:** a montage with no complete pair is `not_applicable`; the reading is sensitivity to value–position correspondence.
- **Anchors:** Section 4.3; Table C.1; Appendix D.1.

B.10 S2.1 Temporal Aperture (D_2)

- **Grid:** apertures of 1, 2, 3, 4 s (Motor), 5, 10, 15, 20, 30 s (Sleep), and 2, 4, 6, 8, 10 s (Emotion, Seizure); centered placement and one random placement per sample; zero or repeat fill outside the aperture.
- **Reference and controls:** full window; repeat fill against zero fill (zero fill removes exterior energy and adds edge steps; repeat fill adds repeated structure); random against centered placement.
- **Quantities:** $R(t)$; t_{90}; $\Delta_{\mathrm{pad}} = \mathrm{AURC}_{\mathrm{repeat}} - \mathrm{AURC}_{\mathrm{zero}}$; $\mathrm{APG} = \mathrm{AURC}_{\mathrm{random}} - \mathrm{AURC}_{\mathrm{center}}$; $\bar{\Delta} = Q(0)[1 - \mathrm{AURC}]$.
- **Applicability:** evaluation-only aperture masking on the locked state; the single random placement gives one realization, not a placement distribution.
- **Anchors:** Section 4.3; Figures 5 and C.2; Table C.4.

B.11 S2.3 Frequency Sensitivity (D_2)

- **Grid:** fourth-order zero-phase Butterworth (`sosfiltfilt`, about 1 s reflection padding); low-pass cutoffs $\{4, 8, 11, 13, 16, 30, 45\}$ Hz and band-stops δ, θ, α, σ, β, γ, μ/β; per-channel RMS renormalization with adapter instance normalization off.
- **Reference and controls:** full spectrum; for each pre-specified target band (Table B.5) the low-pass gain and band-stop collapse form a pair, and the band-stops outside the target band measure the filter's own side effect; the 45 Hz low-pass gives HighBandGap.
- **Quantities:** $R(f) = Q(\mathrm{lp}_f)/Q(\mathrm{full})$; $\mathrm{BandCollapse}(b) = 1 - R(\mathrm{bandstop}_b)$; $\mathrm{HighBandGap} = 1 - R(45)$.
- **Applicability:** the response combines the band's information with the filter's transition band and leakage; it reads dependence under this operator.
- **Anchors:** Section 3.4; Figure D.2; Table C.2.

B.12 S2.5 Alpha Modulation (D_2)

- **Grid:** real-FFT modulation of the 8–13 Hz band with tapered edges; power ratios 0.3, 0.5, 0.7, 1.0, 1.3, 1.5, 2.0; bandwidth ratios 0.5, 0.7, 1.0, 1.3, 1.5; peak shifts from -2 to $+2$ Hz in steps of 0.5 Hz.
- **Reference and controls:** identity coordinate (1.0/1.0/0.0 Hz); the grid is bidirectional around it.
- **Quantities:** $\mathrm{AURC}_{\mathrm{power}}, \mathrm{AURC}_{\mathrm{bw}}, \mathrm{AURC}_{\mathrm{peak}}$; signed axis losses $\bar{\Delta}_a$; MRA.
- **Applicability:** alpha relevance is task-conditional (Table B.5); the edit acts on the spectrum and changes waveform morphology with it.
- **Anchors:** Figure D.3; Table C.1.

B.13 S3.2 Artifact Injection (D_3)

- **Grid:** additive synthetic EMG ($>$ 30 Hz on peripheral frontotemporal channels) and EOG ($<$ 4 Hz drift and blinks on anterior frontal channels), mixed before instance normalization at signal-to-artifact power ratios $\{10, 5, 2, 1, 0.5\}$ (ratios, not decibels); reference ratio 2.
- **Reference and controls:** clean predictions; the per-type dose curve and the EMG–EOG difference are kept separately.

- **Quantities:** $\Delta_a = Q_{\text{clean}} - Q_a(\text{SNR}_{\text{ref}})$; $\text{CAD} = |\Delta_{\text{EMG}} - \Delta_{\text{EOG}}|$; $\text{AUC}_a^{\log}$ (Table B.6), the scorecard quantity.
- **Applicability:** a task without the target channels is `not_applicable`; SNR is matched by mean target-channel power within a type, and the two types use different channel sets, so CAD includes a channel-count difference.
- **Anchors:** Figure D.4; Table C.1.

B.14 S3.3 Background Crossfade (D_3)

- **Grid:** backgrounds RMS-matched per channel; $x_{\text{mix}}(\alpha) = (1-\alpha)x + \alpha n$ at $\alpha \in \{0.0, 0.1, \ldots, 1.0\}$; no post-mix renormalization, adapter instance normalization off.
- **Reference and controls:** $\alpha = 0$; pink noise is the headline background, with white noise, phase-scrambled pink noise, and energy-only attenuation $x' = (1-\alpha)x$ as spectrum, phase, and energy controls.
- **Quantities:** EET, the first descending interpolated crossing of $Q(\alpha) = Q_{\text{ch}} + 0.5L_0$, coded 0 or 1 when the scan starts below or never crosses; $\bar{\Delta}_{\text{pink}}$; SpectrumGap, PhaseGap, EnergyConfoundGap.
- **Anchors:** Section 4.3; Figures 6 and C.3; Table C.7.

B.15 S3.4 Rejection Entropy (D_3)

- **Grid:** the input is replaced by Gaussian thermal noise, synthetic ECG (QRS spikes), or synthetic PPG (pulsatile waveform) under the task's channel identities.
- **Reference and controls:** `original_eeg` is a metadata label; there is no clean-EEG entropy reference and no task score.
- **Quantities:** for covariance eigenvalue proportions p_i over the exposed feature dimension d_h, $H_{\text{norm}} = (-\sum_i p_i \log p_i)/\log d_h \in [0, 1]$.
- **Applicability:** a descriptive dispersion statistic of the representation; not a rejection score.
- **Anchors:** Figure D.5; Table C.7.

B.16 Confirmation Tests for Component-Level Attribution

The profiles compare whole routes. Table B.9 lists the within-backbone manipulations that would turn a cross-model hypothesis (Section 5.2) into a component-level attribution.

Table B.9: **Within-backbone manipulations and the reading that would strengthen an attribution.** A clean-score gain alone attributes nothing.

Manipulation (one backbone, fixed data and capacity)	**Strengthening reading**
Spatial pooling against coordinate encoding (mean pooling, identity flattening, coordinate conditioning)	A reproducible S1.1 / S1.2 change that persists after conditioning on L_0
Context length against boundary policy (zero, repeat, mirror) under a fixed input representation	The context effect recurs across boundary operators
Time-domain patches against STFT spectrograms under one pretraining recipe and a matched sequence length	Predicted S2.3 / S2.5 / S3.2 shifts replicate across seeds without a loss of clean observability
Fixed scaling against per-channel normalization at the adapter, and against a gain-invariant objective, under one recipe	The S3.3 hold at $\alpha = 1$ and the energy-only loss both vanish while L_0 survives; the S2.3 reference costs no lift

C Extended Evidence

Main-text anchors: Section 4 reports the readings and Sections 5.1 and 5.3 interpret them; this appendix gives the numbers and controls behind them. Uncertainty conventions are in Appendix B.7.

This appendix holds the four task scorecards (Appendix C.1), the grids and controls behind the three readings of Section 4.3 (Appendix C.2), one interface note per model (Appendix C.3), and the claim–evidence table (Appendix C.4); Appendix D holds the five other protocols. Nothing here is averaged across tasks into a rank.

C.1 Task Scorecards

Each scorecard fixes one task and lists the coordinates of the seven task-conditioned protocols for the five models (Tables C.1–C.4); S3.4 has no task axis and is reported by source in Appendix D.5. Four conventions hold. (i) Values are mean ± sample standard deviation over the three Stage I seeds (20260528, 20260612, 20260718), with L_0 as the three-seed mean; for S1.1 the region chosen on seed 20260528 is held fixed for the other two seeds. (ii) `N/A` marks a condition the montage does not carry, never a zero response. (iii) S2.1 and S2.3 use clipped AURCs and truncated losses; S2.5 and S3.3 keep raw AURCs and signed losses. (iv) S2.3 and S3.3 change the adapter's normalization (Table B.1; its cost is in Table D.1), so each carries its own local L_0 row; a small loss under a near-zero L_0 reads as absence of lift, not as robustness.

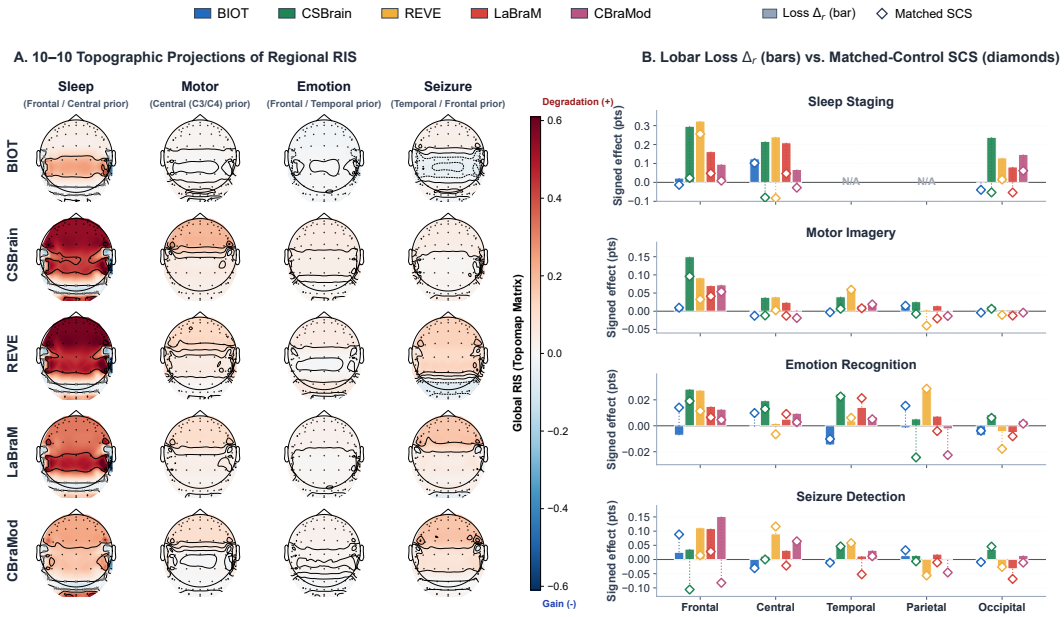


Figure C.1: **Complete region-criticality atlas at the 100% zero-masking endpoint** (S1.1; full-scope version of Figure 4). (A) Model-by-task 10–10 projections of regionwise RIS, each task column headed by its pre-specified target region; (B) regionwise endpoint loss Δ_r bars, each linked by a dotted stem to the SCS diamond of its equal-channel control, in metric points, for all four tasks. Panels visualize seed 20260528; projections summarize lobar endpoint responses and are not source-localization maps.

Table C.1: **Scorecard for Motor Imagery (`physiomi_mi`).** Three Stage I seeds, mean ± sample standard deviation. Losses and $\mathrm{AUC}_a^{\log}$ are in AUROC points; larger values mean larger degradation. S2.3 and S3.3 change the adapter's normalization and are read against their own local L_0.

Protocol	Coordinate	BIOT	CBraMod	CSBrain	LaBraM	REVE
Reference	Clean AUROC	0.511 ± 0.008	0.750 ± 0.002	0.748 ± 0.008	0.678 ± 0.027	0.762 ± 0.003
	Clean lift L_0	0.011	0.250	0.248	0.178	0.262
Spatial (D_1)						
S1.1 Regional masking	Region (seed `20260528`)	parietal	frontal	frontal	frontal	frontal
	Endpoint loss Δ_r	$+0.005 \pm 0.006$	$+0.071 \pm 0.001$	$+0.136 \pm 0.026$	$+0.068 \pm 0.013$	$+0.089 \pm 0.003$
S1.2 Hemispheric swap	Accuracy drop AD	0.000 ± 0.001	0.279 ± 0.002	0.268 ± 0.005	0.178 ± 0.022	0.257 ± 0.015
	Reversal rate PRR	$3.5\% \pm 0.9\%$	$44.3\% \pm 1.1\%$	$39.8\% \pm 6.3\%$	$29.1\% \pm 6.9\%$	$38.3\% \pm 12.2\%$
Temporal and spectral (D_2)						
S2.1 Temporal aperture	Mean loss $\bar{\Delta}$	0.003 ± 0.001	0.124 ± 0.001	0.136 ± 0.010	0.080 ± 0.017	0.124 ± 0.003
S2.3 Frequency sensitivity	Mean loss $\bar{\Delta}$ (log cutoff axis)	0.010 ± 0.004	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.005 ± 0.002
	Local L_0 (RMS reference)	0.011	0.224	0.127	0.133	0.256
S2.5 Alpha modulation	Mean loss $\bar{\Delta}$, power axis	$+0.001 \pm 0.001$	$+0.001 \pm 0.000$	$+0.001 \pm 0.000$	$+0.001 \pm 0.001$	$+0.001 \pm 0.000$
	Mean loss $\bar{\Delta}$, bandwidth axis	$+0.001 \pm 0.001$	$+0.003 \pm 0.000$	$+0.000 \pm 0.001$	$+0.002 \pm 0.001$	$+0.003 \pm 0.000$
	Mean loss $\bar{\Delta}$, peak axis	$+0.003 \pm 0.002$	$+0.003 \pm 0.000$	-0.003 ± 0.001	$+0.001 \pm 0.002$	$+0.002 \pm 0.000$
Artifact and background (D_3)						
S3.2 Artifact injection	EMG $\mathrm{AUC}_a^{\log}$	-0.003 ± 0.005	$+0.013 \pm 0.000$	$+0.005 \pm 0.002$	$+0.035 \pm 0.006$	$+0.007 \pm 0.002$
	EOG $\mathrm{AUC}_a^{\log}$	$+0.000 \pm 0.005$	$+0.023 \pm 0.000$	$+0.012 \pm 0.003$	$+0.020 \pm 0.010$	$+0.018 \pm 0.004$
S3.3 Background crossfade	Event extraction threshold EET	0.151 ± 0.077	0.601 ± 0.002	0.591 ± 0.026	0.573 ± 0.034	0.558 ± 0.014
	Mean loss $\bar{\Delta}$, pink background	0.005 ± 0.005	0.100 ± 0.001	0.102 ± 0.008	0.077 ± 0.014	0.072 ± 0.006
	Local L_0	0.011	0.250	0.248	0.178	0.163

Table C.2: **Scorecard for Sleep Staging (`isruc_sleep`).** Three Stage I seeds, mean ± sample standard deviation. Losses and $\mathrm{AUC}_a^{\log}$ are in balanced accuracy points; larger values mean larger degradation. S2.3 and S3.3 change the adapter's normalization and are read against their own local L_0.

Protocol	Coordinate	BIOT	CBraMod	CSBrain	LaBraM	REVE
Reference	Clean BalAcc	0.637 ± 0.001	0.411 ± 0.004	0.544 ± 0.025	0.498 ± 0.008	0.579 ± 0.024
	Clean lift L_0	0.437	0.211	0.344	0.298	0.379
Spatial (D_1)						
S1.1 Regional masking	Region (seed `20260528`)	central	occipital	frontal	central	frontal
	Endpoint loss Δ_r	$+0.122 \pm 0.007$	$+0.147 \pm 0.003$	$+0.274 \pm 0.044$	$+0.174 \pm 0.037$	$+0.323 \pm 0.070$
S1.2 Hemispheric swap	Accuracy drop AD	$+0.008 \pm 0.001$	-0.001 ± 0.001	$+0.008 \pm 0.008$	-0.001 ± 0.010	$+0.016 \pm 0.015$
Temporal and spectral (D_2)						
S2.1 Temporal aperture	Mean loss $\bar{\Delta}$	0.065 ± 0.003	0.167 ± 0.004	0.189 ± 0.069	0.150 ± 0.008	0.212 ± 0.034
S2.3 Frequency sensitivity	Mean loss $\bar{\Delta}$ (log cutoff axis)	0.320 ± 0.006	0.026 ± 0.002	0.041 ± 0.021	0.051 ± 0.041	0.158 ± 0.024
	Local L_0 (RMS reference)	0.432	0.045	0.081	0.145	0.379
S2.5 Alpha modulation	Mean loss $\bar{\Delta}$, power axis	$+0.019 \pm 0.001$	$+0.004 \pm 0.001$	$+0.009 \pm 0.007$	$+0.003 \pm 0.002$	$+0.009 \pm 0.004$
	Mean loss $\bar{\Delta}$, bandwidth axis	$+0.005 \pm 0.001$	-0.002 ± 0.000	$+0.004 \pm 0.005$	$+0.009 \pm 0.003$	$+0.003 \pm 0.004$
	Mean loss $\bar{\Delta}$, peak axis	$+0.006 \pm 0.001$	$+0.001 \pm 0.001$	$+0.006 \pm 0.003$	$+0.006 \pm 0.001$	$+0.005 \pm 0.007$
Artifact and background (D_3)						
S3.2 Artifact injection	EMG $\mathrm{AUC}_a^{\log}$	`N/A`	`N/A`	`N/A`	`N/A`	`N/A`
	EOG $\mathrm{AUC}_a^{\log}$	$+0.129 \pm 0.008$	$+0.015 \pm 0.002$	$+0.039 \pm 0.030$	$+0.019 \pm 0.011$	$+0.068 \pm 0.016$
S3.3 Background crossfade	Event extraction threshold EET	0.308 ± 0.005	0.313 ± 0.011	0.363 ± 0.005	0.473 ± 0.053	0.394 ± 0.007
	Mean loss $\bar{\Delta}$, pink background	0.299 ± 0.002	0.124 ± 0.004	0.200 ± 0.022	0.154 ± 0.011	0.112 ± 0.014
	Local L_0	0.437	0.211	0.344	0.298	0.187

`N/A`: the 6-channel ISRUC montage has no peripheral temporal electrodes, so EMG injection does not apply.

Table C.3: **Scorecard for Emotion Recognition (`seed_iv_emotion`).** Three Stage I seeds, mean ± sample standard deviation. Losses and $\mathrm{AUC}_a^{\log}$ are in AUROC points; larger values mean larger degradation. S2.3 and S3.3 change the adapter's normalization and are read against their own local L_0.

Protocol	Coordinate	BIOT	CBraMod	CSBrain	LaBraM	REVE
Reference	Clean AUROC	0.497 ± 0.009	0.530 ± 0.001	0.558 ± 0.001	0.547 ± 0.013	0.568 ± 0.014
	Clean lift L_0	-0.003	0.030	0.058	0.047	0.068
Spatial (D_1)						
S1.1 Regional masking	Region (seed `20260528`)	central	frontal	frontal	frontal	parietal
	Endpoint loss Δ_r	$+0.003 \pm 0.002$	$+0.012 \pm 0.001$	$+0.026 \pm 0.006$	$+0.010 \pm 0.008$	$+0.013 \pm 0.013$
S1.2 Hemispheric swap	Accuracy drop AD	-0.004 ± 0.004	$+0.018 \pm 0.001$	$+0.012 \pm 0.002$	$+0.018 \pm 0.010$	$+0.012 \pm 0.006$
Temporal and spectral (D_2)						
S2.1 Temporal aperture	Mean loss $\bar{\Delta}$	0.000 ± 0.000	0.019 ± 0.001	0.015 ± 0.004	0.017 ± 0.002	0.009 ± 0.001
S2.3 Frequency sensitivity	Mean loss $\bar{\Delta}$ (log cutoff axis)	0.011 ± 0.004	0.001 ± 0.000	0.000 ± 0.000	0.003 ± 0.002	0.023 ± 0.009
	Local L_0 (RMS reference)	-0.004	0.013	0.010	0.021	0.067
S2.5 Alpha modulation	Mean loss $\bar{\Delta}$, power axis	$+0.000 \pm 0.000$	$+0.000 \pm 0.000$	$+0.001 \pm 0.000$	$+0.001 \pm 0.001$	$+0.000 \pm 0.000$
	Mean loss $\bar{\Delta}$, bandwidth axis	-0.001 ± 0.001	$+0.002 \pm 0.000$	$+0.000 \pm 0.000$	$+0.001 \pm 0.000$	$+0.000 \pm 0.001$
	Mean loss $\bar{\Delta}$, peak axis	$+0.004 \pm 0.001$	$+0.001 \pm 0.000$	-0.002 ± 0.001	$+0.000 \pm 0.003$	$+0.000 \pm 0.001$
Artifact and background (D_3)						
S3.2 Artifact injection	EMG $\mathrm{AUC}_a^{\log}$	-0.017 ± 0.013	$+0.009 \pm 0.000$	-0.001 ± 0.000	$+0.025 \pm 0.008$	$+0.002 \pm 0.005$
	EOG $\mathrm{AUC}_a^{\log}$	-0.024 ± 0.008	-0.001 ± 0.001	$+0.012 \pm 0.002$	$+0.006 \pm 0.002$	$+0.003 \pm 0.003$
S3.3 Background crossfade	Event extraction threshold EET	0.231 ± 0.399	0.582 ± 0.259	0.663 ± 0.030	0.475 ± 0.109	0.103 ± 0.178
	Mean loss $\bar{\Delta}$, pink background	-0.003 ± 0.008	0.013 ± 0.002	0.018 ± 0.002	0.024 ± 0.010	0.005 ± 0.009
	Local L_0	-0.003	0.030	0.058	0.047	-0.003

Table C.4: **Scorecard for Seizure Detection (`siena_seiz`).** Three Stage I seeds, mean ± sample standard deviation. Losses and $\mathrm{AUC}_a^{\log}$ are in AUROC points; larger values mean larger degradation. S2.3 and S3.3 change the adapter's normalization and are read against their own local L_0.

Protocol	Coordinate	BIOT	CBraMod	CSBrain	LaBraM	REVE
Reference	Clean AUROC	0.811 ± 0.013	0.887 ± 0.003	0.870 ± 0.020	0.652 ± 0.029	0.844 ± 0.054
	Clean lift L_0	0.311	0.387	0.370	0.152	0.344
Spatial (D_1)						
S1.1 Regional masking	Region (seed `20260528`)	frontal	frontal	temporal	frontal	frontal
	Endpoint loss Δ_r	$+0.025 \pm 0.017$	$+0.126 \pm 0.022$	$+0.030 \pm 0.022$	$+0.091 \pm 0.036$	$+0.130 \pm 0.017$
S1.2 Hemispheric swap	Accuracy drop AD	$+0.002 \pm 0.005$	$+0.078 \pm 0.008$	$+0.003 \pm 0.001$	$+0.017 \pm 0.030$	$+0.036 \pm 0.031$
Temporal and spectral (D_2)						
S2.1 Temporal aperture	Mean loss $\bar{\Delta}$	0.005 ± 0.001	0.453 ± 0.009	0.047 ± 0.030	0.019 ± 0.033	0.055 ± 0.012
S2.3 Frequency sensitivity	Mean loss $\bar{\Delta}$ (log cutoff axis)	0.184 ± 0.023	0.113 ± 0.003	0.074 ± 0.021	0.065 ± 0.014	0.062 ± 0.022
	Local L_0 (RMS reference)	0.304	0.064	0.022	0.098	0.348
S2.5 Alpha modulation	Mean loss $\bar{\Delta}$, power axis	-0.003 ± 0.001	$+0.000 \pm 0.000$	$+0.000 \pm 0.000$	$+0.004 \pm 0.007$	$+0.001 \pm 0.001$
	Mean loss $\bar{\Delta}$, bandwidth axis	-0.010 ± 0.004	$+0.004 \pm 0.000$	$+0.001 \pm 0.002$	$+0.011 \pm 0.008$	$+0.012 \pm 0.003$
	Mean loss $\bar{\Delta}$, peak axis	$+0.001 \pm 0.004$	$+0.005 \pm 0.000$	$+0.001 \pm 0.002$	$+0.011 \pm 0.007$	$+0.007 \pm 0.003$
Artifact and background (D_3)						
S3.2 Artifact injection	EMG $\mathrm{AUC}_a^{\log}$	$+0.042 \pm 0.008$	-0.014 ± 0.003	$+0.032 \pm 0.008$	-0.002 ± 0.013	$+0.007 \pm 0.009$
	EOG $\mathrm{AUC}_a^{\log}$	$+0.069 \pm 0.017$	$+0.001 \pm 0.000$	$+0.022 \pm 0.005$	$+0.040 \pm 0.026$	$+0.047 \pm 0.013$
S3.3 Background crossfade	Event extraction threshold EET	0.490 ± 0.063	1.000 ± 0.000	1.000 ± 0.000	0.417 ± 0.291	0.153 ± 0.133
	Mean loss $\bar{\Delta}$, pink background	0.165 ± 0.007	0.002 ± 0.001	-0.016 ± 0.007	0.074 ± 0.030	0.051 ± 0.104
	Local L_0	0.311	0.387	0.370	0.152	0.058

C.2 Behind the Three Readings

Each paragraph carries the definitions and controls that Section 4.3 reads but does not print; the figures draw seed 20260528, and every number below is a three-seed value unless marked.

Spatial: a regional loss is not regional specificity. Figure C.1 gives the four-task version of the Motor/Seizure strip: the endpoint loss Δ_r of every region the montage carries beside the SCS of its equal-channel control. The operational atlas assigns `FC*` to the frontal group and the five groups differ in size (Table B.8), so a comparison *between* regions carries the atlas and the region size with it; the Motor contrast is read within one frontal group ($n_r = 24/64$) against its own equal-channel control, the comparison that control was pre-specified to license. Over the $1{,}080$ regional/scattered pairs of the zero-masking grid, 3 seeds $\times$ 5 models $\times$ 18 task–regions $\times$ 4 masking ratios, 206 have a regional cost above 0.05 and 66 of those carry an equal-channel cost at least as large; the median SCS over the 872 pairs whose regional cost lies within ± 0.05 is -0.0004. Zeroing and the power-conserving temporal shuffle of the same channels (Appendix B.8) agree on Motor (every SCS within 0.04) but not on Sleep and Seizure, where 9 of the 90 cells differ by 0.10 or more and 6 of those reverse sign, so an SCS is reported with its operator and a sign that depends on the operator is not read as specific. On Sleep, where a region is two of six channels, REVE's frontal SCS is 0.182 ± 0.121 under zeroing and -0.094 ± 0.016 under shuffling, and CSBrain's is 0.030 ± 0.006 and 0.177 ± 0.055; on Seizure, CBraMod's frontal cell (RIS 0.142 ± 0.025) has SCS -0.132 ± 0.044 under zeroing and 0.035 ± 0.005 under shuffling (Figure C.1B, Seizure row). None of the three is read as specific. BIOT's 18-derivation template has four live derivations on ISRUC (F3–C3, F4–C4, C3, C4; O1/O2 enter none), so its Sleep occipital cell is 0.000 by construction.

Temporal: a zero-fill loss is read beside its repeat-fill control. Figure C.2 shows the complete aperture grid with its retention curves, clipped response losses, and padding audit. At the shortest aperture, CBraMod's Seizure 2 s window retains 0.081 ± 0.002 of the full-window score under zero filling and 0.985 ± 0.001 under repeat filling (0.958 ± 0.002 at a random position); over the whole grid the exchange recovers 0.441 ± 0.008 metric points, $97.3\% \pm 0.1\%$ of the integrated loss 0.453 ± 0.009. The exchange is that large for CBraMod alone on Seizure ($\Delta_{\mathrm{pad}} = \mathrm{AURC}_{\mathrm{repeat}} - \mathrm{AURC}_{\mathrm{zero}}$ of $+0.497 \pm 0.008$ against -0.046 to $+0.049$ for the other four); on Sleep every route pays a zero-fill cost ($+0.087$ to $+0.361$) and repeat fill returns 85–93% of each route's integrated zero-fill loss there, $(\mathrm{AURC}_{\mathrm{repeat}} - \mathrm{AURC}_{\mathrm{zero}})/(1 - \mathrm{AURC}_{\mathrm{zero}})$ of 0.854 ± 0.007, 0.887 ± 0.007, 0.855 ± 0.060, 0.919 ± 0.024, and 0.925 ± 0.026 for BIOT, CBraMod, CSBrain, LaBraM, and REVE, and on Motor repeat fill is the worse policy (-0.050 to -0.019), as on the planted stationary codes of Appendix E.3. A zero-fill loss is therefore read beside its repeat-fill control, and the recovery is a property of the fixed-shape input interface, read on one model, one task, and one cohort.

Background: CBraMod and CSBrain follow input amplitude; BIOT and REVE do not. The background is RMS-matched per channel to the recording before mixing and the mixture is not renormalized (Appendix B.14), so at $\alpha = 1$ the input carries the recording's per-channel amplitude and none of its waveform, while the energy-only control $(1 - \alpha)x$ removes that amplitude and keeps the waveform. Table C.5 gives the Seizure endpoints: at $\alpha = 1$ CBraMod and CSBrain score 0.905 ± 0.003 and 0.894 ± 0.018 on the pink background and 0.869–0.902 under the white and phase-scrambled backgrounds, while the other three routes are at chance, and under attenuation alone the two holds invert (0.264 ± 0.010 and 0.323 ± 0.125 at $\alpha = 0.9$) while BIOT, which scales every derivation by its own 95th percentile, stays at 0.811 ± 0.013. On Sleep and Motor every route is at chance at $\alpha = 1$; the energy-only losses on Sleep are 0.096–0.148 for the three fixed-scale routes against 0.022 (BIOT) and -0.032 (REVE), and at most 0.025 on Motor. On Sleep the energy-only control at $\alpha = 0.9$ takes the three fixed-scale routes from 0.411, 0.544, and 0.498 to 0.247 ± 0.001, 0.342 ± 0.042, and 0.331 ± 0.018, 56–78% of their lift above 0.2, while BIOT stays at 0.636 ± 0.001 and REVE moves from its local 0.387 to 0.463 ± 0.050. REVE's local lifts here (0.058, 0.187, 0.163 on Seizure, Sleep, Motor) record that switching its z-scoring off moves it from the scale it was fitted on. Read with the cost of the S2.3 reference (Table D.1), the predictions of the three fixed-scale routes follow input amplitude on Sleep, and those of CBraMod and CSBrain also on Seizure; BIOT and REVE do not. This is a statement about the input each readout sees, not about the representation. Figure C.3 draws the sweeps for all four tasks.

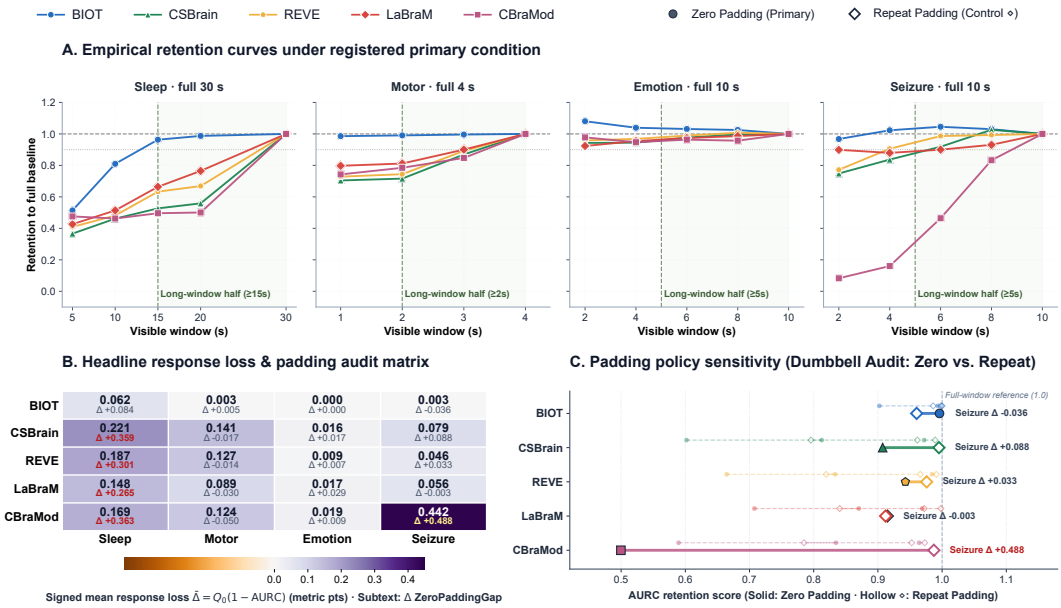


Figure C.2: **Temporal-aperture response and padding audit** (S2.1; full-grid version of Figure 5, deferred from Section 4.3). (A) Retention curves over the center-plus-zero-fill aperture grid; (B) signed mean response loss $\bar{\Delta}$ and repeat-versus-zero padding gap; (C) padding-policy sensitivity, zero padding (solid) against repeat padding (hollow ⋄), with CBraMod's $+0.488$ gap on Seizure. Panels visualize seed 20260528.

Table C.5: **Seizure endpoints of the background mixture (S3.3).** AUROC on pure background ($\alpha = 1$) under the three backgrounds, and under attenuation alone at $\alpha = 0.9$; signed mean losses in AUROC points ($N_{\text{seed}} = 3$; chance 0.5).

Model	L_0	**Pink** α=1	**White** α=1	**Phase-scr.** α=1	**Energy-only** α=0.9	$\bar{\Delta}_{\text{energy}}$
BIOT	0.311	0.485±0.012	0.507±0.009	0.478±0.011	0.811±0.013	+0.015±0.001
CBraMod	0.387	0.905±0.003	0.902±0.002	0.900±0.005	0.264±0.010	+0.135±0.003
CSBrain	0.370	0.894±0.018	0.869±0.039	0.896±0.017	0.323±0.125	+0.107±0.036
LaBraM	0.152	0.518±0.009	0.473±0.020	0.509±0.033	0.773±0.052	-0.029±0.022
REVE	0.058	0.522±0.089	0.494±0.064	0.532±0.148	0.685±0.049	+0.010±0.036

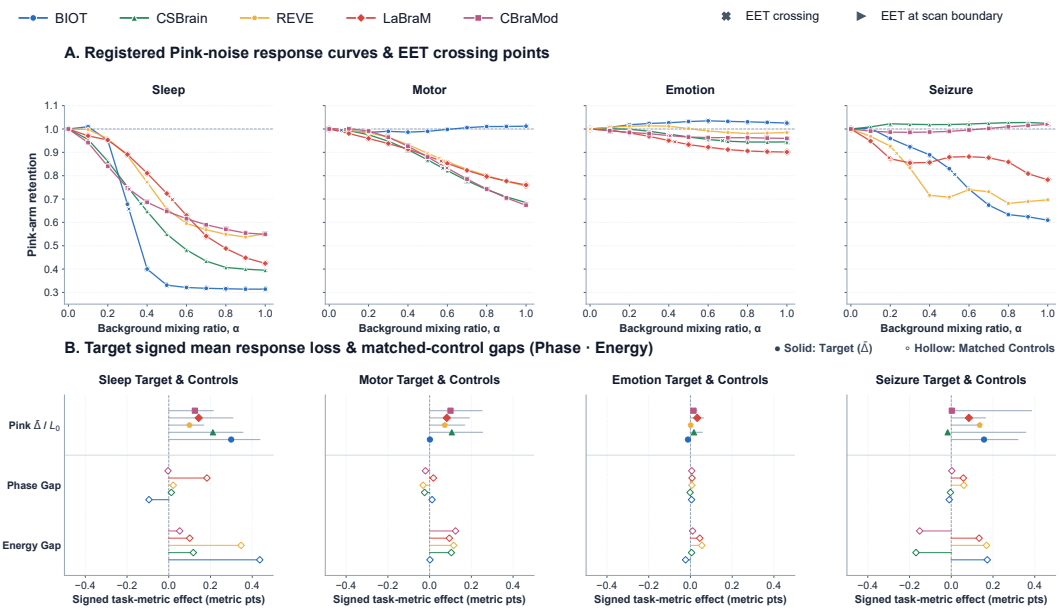


Figure C.3: **Background crossfade response** (S3.3; four-task version of Figure 6, deferred from Section 4.3). (A) Pink-noise retention curves per task, with EET crossings and a scan-boundary glyph wherever the half-lift boundary is never reached; (B) per-task contrasts for the pink-noise signed mean response loss $\bar{\Delta}_{\text{pink}}$, the phase-control gap, and the energy-control gap, each beside its local L_0. Panels visualize seed 20260528.

C.3 One Interface Note per Model

Table C.6 records, per route, the interface facts that fix how its readings are read. They are read from the released model code and the adapters of Appendix A.3; they say what reaches the backbone and in what arrangement, not what the backbone does with it.

Table C.6: **One interface note per model.**

Model	Interface facts and the readings they fix
BIOT	Each of 18 bipolar derivations is divided by the 95th percentile of its own $\lvert x\rvert$ before the STFT, so a uniform gain per derivation leaves the input unchanged and attenuation cannot move it. Its input is 1 s STFT magnitudes, so a band-stop edits the input features directly. A slot-indexed channel embedding is added to each derivation's tokens before all tokens are averaged. Missing derivations are zero-filled: 4 of 18 are live on ISRUC, 18 of 18 elsewhere.
CBraMod	Fixed $\times 0.01$, so amplitude reaches the backbone up to a constant. Position enters through a depthwise convolution over the channel–patch grid in a fixed 10–10 order, so moving a channel changes its embedding; attention alternates between the channel axis and the patch axis.
CSBrain	Fixed $\times 0.01$. Channels are sorted by scalp region before entry, and region grouping is part of the interface; its multi-scale kernels (1, 3, 5) act along the 1 s patch sequence and across the channels of a region, not inside the waveform.
LaBraM	Fixed $\times 0.01$. Channels are embedded by their standard 10–20 index; all tokens of the window enter one transformer with the released temporal-position table interpolated to the window length; the deployed backbone carries no quantizer. Amplitude reaches the backbone, and the Seizure prediction does not follow it (energy-only loss -0.029 ± 0.022).
REVE	Per-channel z-score, so gain and offset are removed before the backbone: the S2.3 renormalization leaves it unchanged, and switching the z-score off in S3.3 lowers its local lifts. Position is a 4D Fourier embedding of the sensor's 3D coordinates and the patch time, so a left–right swap is a sign change of one coordinate.

C.4 Claim–Evidence Table

Table C.7 lists each main-text claim with the measurements that carry it and the controls and uncertainty it is read with. Additions that would strengthen the rows: subject-cluster inference for S1.2, subject resampling propagated through response surfaces, repeated scattered controls, and more than one Stage II realization.

Table C.7: **Claims, the evidence that carries them, and the controls and uncertainty they are read with.**

Claim	Evidence	Controls and uncertainty
Clean scores conceal diagnostic differences	CBraMod and CSBrain Motor AUROCs (0.750 ± 0.002 vs 0.748 ± 0.008) coexist with frontal S1.1 losses (0.071 ± 0.001 vs 0.136 ± 0.026) and SCS (0.052 ± 0.001 vs 0.085 ± 0.018).	Clean L_0; the same Motor route shape, parameter count, and $1/100$ input scaling for the two models; full regional and scattered profiles; frontal size $n_r = 24/64$. Clean scores carry subject-bootstrap intervals; S1.1 coordinates are per-seed point estimates summarized over $N_{\text{seed}} = 3$.
Regional impact $\neq$ regional specificity	REVE and CSBrain Sleep frontal RIS (0.555 ± 0.106 vs 0.502 ± 0.060) coexist with SCS (0.182 ± 0.121 vs 0.030 ± 0.006); 66 of the 206 pairs with a regional cost above 0.05 carry an equal-channel cost at least as large.	Equal-channel scattered masks; zero-versus-shuffle contrast, which reverses sign in 6 of the 9 cells where the two operators differ by 0.10 or more; one seeded scattered subset per ratio; per-seed RIS/SCS summarized over $N_{\text{seed}} = 3$.
The fill policy carries the loss	CBraMod Seizure $\bar{\Delta} = 0.453 \pm 0.009$ comes with a repeat-versus-zero padding contrast of 0.441 ± 0.008 metric points, 97.3% of that loss; the 2 s zero-filled window retains 0.081 ± 0.002 against 0.985 ± 0.001 under repeat fill.	Full-window reference; both fill policies; aperture-position controls; on Motor the same exchange costs 0.019–0.050; $N_{\text{seed}} = 3$; Siena has $n_{\text{test}} = 3$ subjects.
Response profiles are task-conditional	CBraMod S2.1 losses are 0.124 ± 0.001 on Motor and 0.453 ± 0.009 on Seizure, with different control gaps.	Task-specific clean metrics and L_0; physical window and grid; padding and position controls; one source per task; Siena $n_{\text{test}} = 3$.
Little lift, no reading	BIOT Motor has $L_0 = 0.011$ with near-zero perturbation losses; Emotion lifts range from -0.003 to 0.068.	Clean metric, chance level, local L_0, absolute task-metric responses; Emotion $n_{\text{test}} = 3$.
Amplitude reaches three routes and not two	S2.3 reference cost on Seizure $0.887 \to 0.564$, $0.870 \to 0.522$, $0.652 \to 0.598$ against $0.811 \to 0.804$ and $0.844 \to 0.848$; pure background at $\alpha = 1$ scores 0.905 ± 0.003 and 0.894 ± 0.018; energy-only losses 0.135 ± 0.003 and 0.107 ± 0.036 against 0.015 ± 0.001 for BIOT.	Declared amplitude transforms (Table A.2) and the normalization settings of Table B.1; local L_0 per protocol; Motor as the task on which no route follows amplitude; read on the input, not on the representation.
S3.4 reads dispersion, not rejection	Normalized covariance-spectrum entropy H_{norm} under thermal noise, ECG, and PPG.	Source identity, feature dimension d_h, frozen channel identities; deterministic point estimates; no clean-EEG entropy reference.

D Protocol Results Deferred from the Main Text

Main-text anchor: Section 4 reads S1.2, S2.3, and S2.5 in part and does not report S3.2 or S3.4; this appendix gives all five in full.

The five protocols ran under the specifications of Appendix B. Figures use seed 20260528; numbers are three-seed values ($N_{\text{seed}} = 3$, mean $\pm$ sample standard deviation) unless marked, with the per-protocol coordinates in the scorecards of Appendix C.1.

D.1 Hemispheric Asymmetry (S1.2)

On Motor Imagery, CBraMod and CSBrain show the largest drops ($\text{AD} = 0.279 \pm 0.002$ and 0.268 ± 0.005) and the highest reversal rates ($\text{PRR} = 44.3\% \pm 1.1\%$ and $39.8\% \pm 6.3\%$), followed by REVE (0.257 ± 0.015; $38.3\% \pm 12.2\%$) and LaBraM (0.178 ± 0.022; $29.1\% \pm 6.9\%$); on seed 20260528 the class-stratified paired bootstrap intervals of AD ($B = 999$) are $[0.257, 0.302]$, $[0.253, 0.292]$, $[0.224, 0.261]$, and $[0.158, 0.198]$. On Sleep, the pre-specified negative-control task, AD stays within ± 0.016 for all five models; on Emotion within 0.018; on Seizure within 0.036 except CBraMod ($+0.078 \pm 0.008$; three test subjects). BIOT's Motor change is $\text{AD} = 0.000 \pm 0.001$ (95% CI $[-0.008, 0.009]$; $\text{PRR} = 3.5\% \pm 0.9\%$) on a clean AUROC of 0.511 ± 0.008 ($L_0 = 0.011$), so its row records absence of lift rather than absence of lateral structure. The swap moves no content and only exchanges the positions of the complete homologous pairs a montage carries, so the reading is sensitivity to the value–position correspondence at the interface (Table C.6), not a lateralization mechanism; the readout refit against this reading is in Appendix F.1.

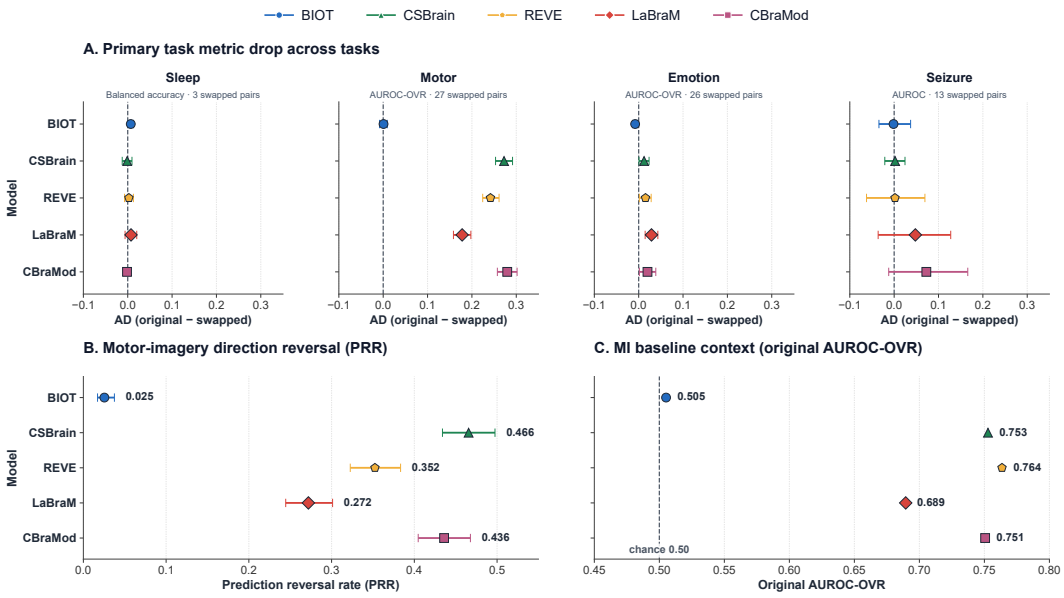


Figure D.1: **Hemispheric asymmetry** (S1.2). (A) Accuracy drop (AD) after left–right channel exchange, with sensitivity on Motor Imagery and near-zero change on the non-lateralized tasks; (B) Motor-imagery reversal rate (PRR) on true left/right trials with Wilson 95% intervals; (C) clean baseline context. Panels visualize seed 20260528.

D.2 Frequency Sensitivity (S2.3)

S2.3 replaces the adapter's z-scoring by a per-channel RMS renormalization applied to every filtered condition and to the full-spectrum reference (Table B.1), so every band reading is taken against that reference, and Table D.1 gives what the reference itself costs. BIOT and REVE do not move, whereas the three fixed-scale routes lose a substantial part of their Sleep and Seizure lift before any filter is applied (on Seizure, CBraMod $0.887 \to 0.564$ and CSBrain $0.870 \to 0.522$ lose most of theirs, and LaBraM $0.652 \to 0.598$ about a third), so their band readings on those tasks are taken at local lifts of 0.022–0.145 and carry little. Figure D.2 keeps the two views apart: cumulative low-pass retention over the seven cutoffs with the reference drawn separately, and the signed collapse under each band-stop, with the pre-specified target bands of Table B.5 marked. For BIOT, whose input features are the STFT magnitudes themselves, the δ band-stop takes Sleep to chance ($0.632 \to 0.209 \pm 0.002$) and low-pass retention returns to 1 only at 45 Hz; on Seizure its retention peaks at the 13 Hz cutoff

(0.965 ± 0.024) and falls back to 0.809 ± 0.025 at 45 Hz. On Sleep its β and σ band-stops collapse 0.411 ± 0.009 and 0.188 ± 0.010 over three seeds, against 0.167 ± 0.027 and 0.058 ± 0.020 for REVE. REVE, the other route with a readable Sleep frequency response, falls to 0.230 ± 0.050 under the δ stop. On Motor the four above-chance routes are unmoved by the α (mu) and β stops (collapse within 0.02) and lose 0.06–0.15 of their reference under the δ stop. The filters are fourth-order zero-phase Butterworth stages, so a band-stop attenuates the band rather than removing it, and a collapse reads dependence on the band under this operator, transition bands and leakage included (Appendix B.11).

Table D.1: **Cost of the S2.3 reference.** Clean metric → full-spectrum reference under per-channel RMS renormalization with adapter z-scoring off ($N_{\text{seed}} = 3$). Chance is 0.5 (AUROC) and 0.2 (BalAcc).

Task	BIOT	CBraMod	CSBrain	LaBraM	REVE
Motor (AUROC)	0.511→0.511±0.005	0.750→0.724±0.001	0.748→0.627±0.011	0.678→0.633±0.022	0.762→0.756±0.003
Sleep (BalAcc)	0.637→0.632±0.002	0.411→0.245±0.000	0.544→0.281±0.019	0.498→0.345±0.079	0.579→0.579±0.025
Emotion (AUROC)	0.497→0.496±0.007	0.530→0.513±0.001	0.558→0.510±0.005	0.547→0.521±0.006	0.568→0.567±0.016
Seizure (AUROC)	0.811→0.804±0.012	0.887→0.564±0.002	0.870→0.522±0.011	0.652→0.598±0.053	0.844→0.848±0.047

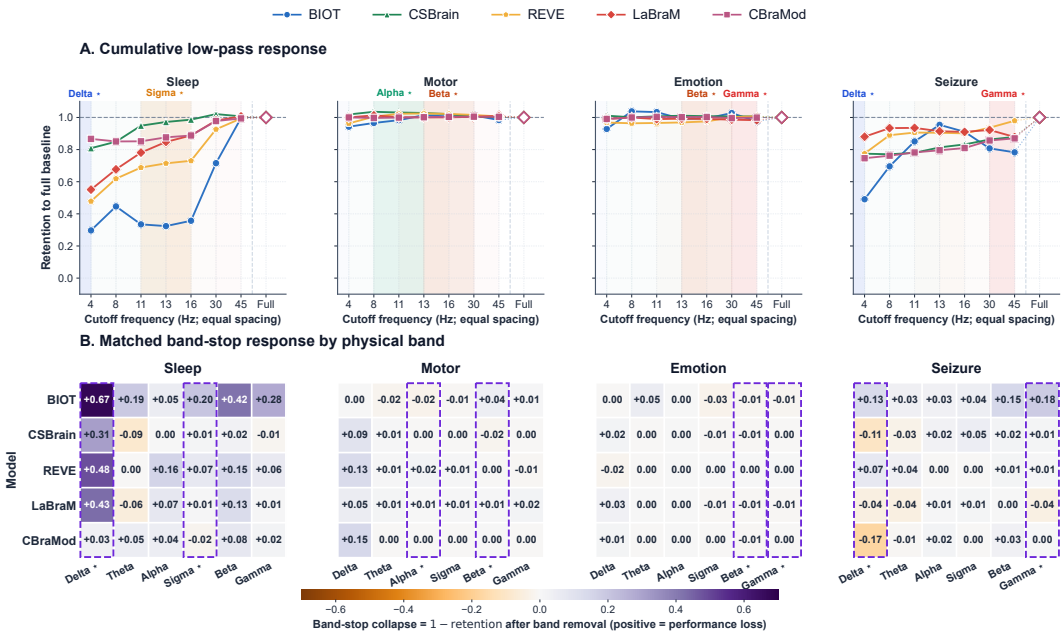


Figure D.2: **Cumulative low-pass and band-stop profile** (S2.3). Low-pass retention over seven cutoffs (4–45 Hz) and band-stop collapse matrices for the four tasks; the full-spectrum reference is drawn separately. Panels visualize seed 20260528.

D.3 Alpha Modulation (S2.5)

S2.5 edits the 8–13 Hz band of the real FFT along three axes around the identity coordinate, with the power axis applied as an amplitude gain of $\sqrt{s}$ so that s is the band-power ratio (Appendix B.12). Figure D.3 gives the response along the three axes. Every axis loss in the scorecards is within 0.019 metric points: on Sleep and Seizure, at lifts of 0.15–0.44, the largest are BIOT's power-axis loss on Sleep ($+0.019 \pm 0.001$) and LaBraM's and REVE's bandwidth losses on Seizure ($+0.011 \pm 0.008$ and $+0.012 \pm 0.003$); on Motor and Emotion every axis is within 0.005. Within this range the predictions do not depend on the alpha microstructure, which agrees with the α band-stop on Motor; the edit acts on the spectrum and changes the waveform with it, and the reading is a null response under this operator, not a statement about what the backbone extracts from the band.

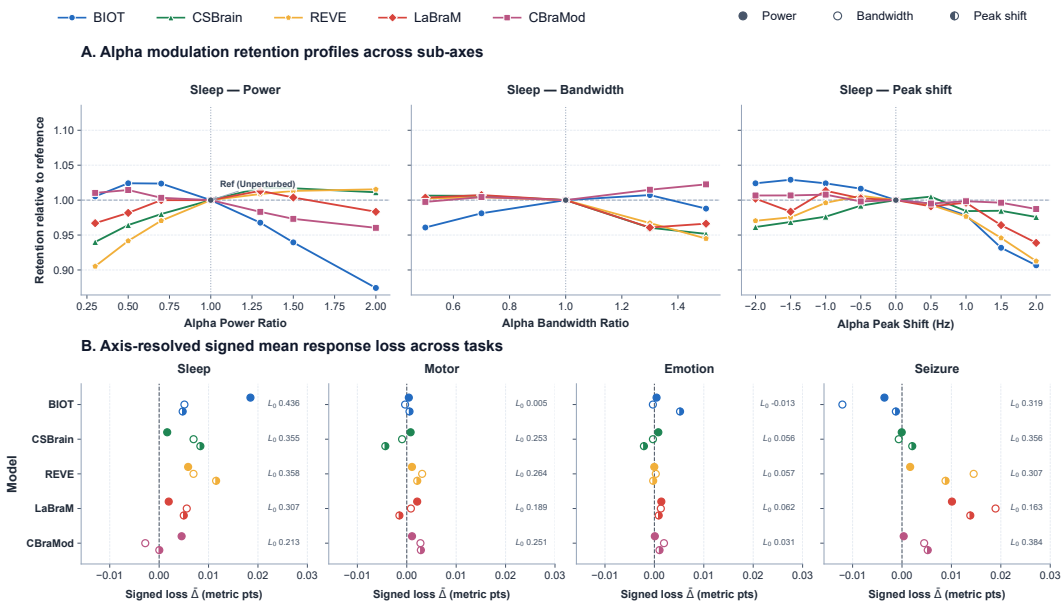


Figure D.3: **Alpha modulation response** (S2.5). (A) Sleep retention along the power, bandwidth, and peak-shift axes, centered on the identity coordinate; (B) axis-resolved signed mean response loss $\bar{\Delta}_a$ for all four tasks beside the local L_0. The three axes are shown separately. Panels visualize seed 20260528.

D.4 Artifact Injection (S3.2)

S3.2 adds synthetic EOG and EMG waveforms in μV, before the adapter's amplitude transform, at signal-to-artifact ratios $\{10, 5, 2, 1, 0.5\}$; EOG is written onto the anterior-frontal channels a montage carries and EMG onto the peripheral fronto-temporal ones, so on ISRUC EOG lands on F3 and F4, two of BIOT's four live derivations, and there is no EMG target (Appendix B.13). Figure D.4 gives the dose–response curves. At the most severe dose, EOG takes BIOT's Sleep score from 0.637 to 0.362 ± 0.015 ($\mathrm{AUC}_a^{\log}$ 0.129 ± 0.008) and REVE's to 0.452 ± 0.030 (0.068 ± 0.016), while CBraMod moves from 0.411 to 0.378 ± 0.002 (0.015 ± 0.002). On Seizure, EOG costs BIOT, REVE, and LaBraM 0.115, 0.063, and 0.030 metric points at that dose, EMG costs CSBrain 0.076, and CBraMod moves by at most 0.007 under either source at a lift of 0.387; under EOG the Seizure response of CBraMod and CSBrain is a null ($\mathrm{AUC}_a^{\log}$ 0.001 and 0.022 ± 0.005). Motor and Emotion move by at most 0.052 and 0.030. The refit of the Seizure readout against both sources is in Appendix F.2, which reports retention rather than $\mathrm{AUC}_a^{\log}$.

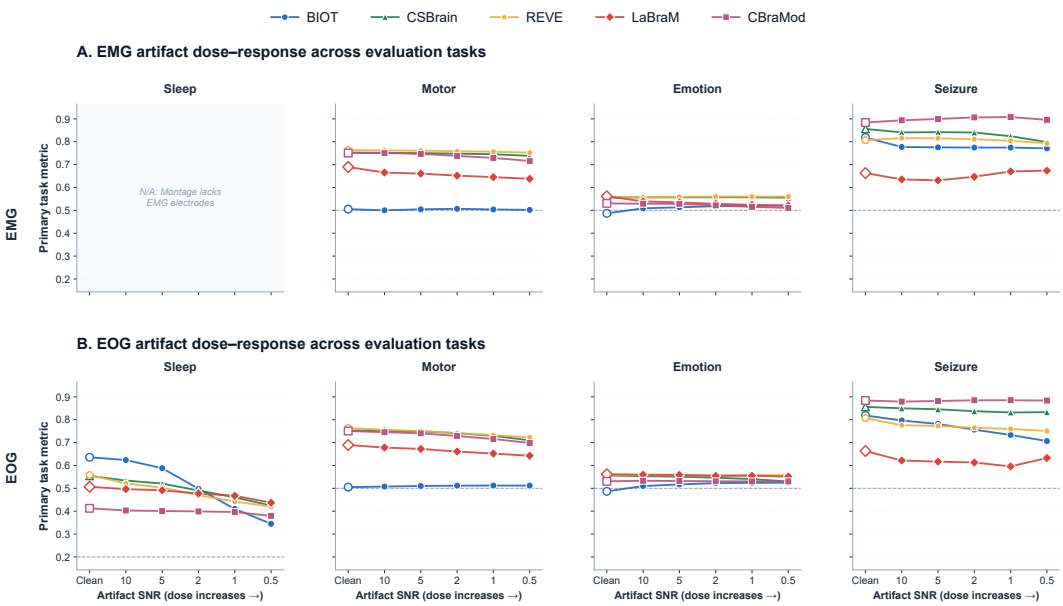


Figure D.4: **Artifact dose–response curves** (S3.2). Primary metric across decreasing signal-to-artifact ratio (clean to 0.5) for EMG and EOG, with the chance level beside each panel. `N/A`: montage without the target channels (ISRUC lacks peripheral temporal EMG sites). Panels visualize seed 20260528.

D.5 Rejection Entropy (S3.4)

S3.4 replaces the whole input, after the adapter's amplitude transform, by Gaussian thermal noise, synthetic ECG, or synthetic PPG under the task's channel identities, and collects the backbone output over 18,702 windows drawn from the four sources. The protocol mean-pools each route's output sequence to one vector per window before the covariance is formed, so BIOT's internally pooled vector ($d_h = 256$) and the pooled streams of the other four routes ($d_h = 200, 200, 200, 512$) are compared on the same kind of object, and $H_{\text{norm}} = H/\log d_h$ is the normalized entropy of the covariance eigenvalues. The three seeds give identical values: the backbone is frozen, the readout does not enter, and the inputs are synthesized deterministically. Figure D.5 gives H_{norm} per model and source: ECG gives the most dispersed spectrum for the four patch-based routes (0.258–0.398) and the least for BIOT (0.033), whose three values lie within 0.033–0.041. Higher values are more dispersed covariance spectra and lower values more concentrated ones; no clean-EEG reference was acquired, no composite across sources is formed, and the reading is a dispersion statistic of the exposed representation, not a rejection score (Table C.7).

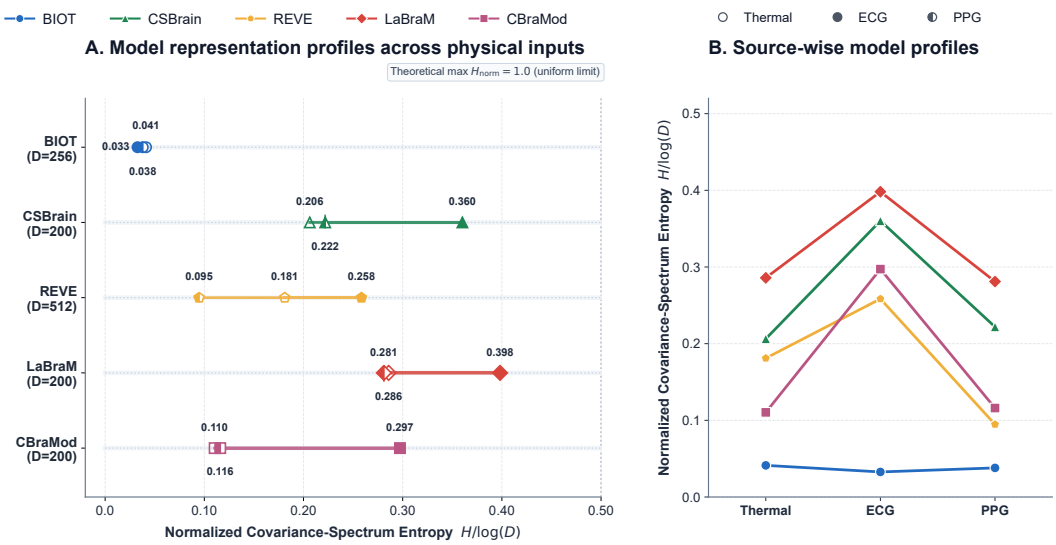


Figure D.5: **Rejection entropy by source** (S3.4). (A) Within-model covariance dispersion across thermal noise, ECG, and PPG, $H_{\text{norm}} = H/\log(d_h)$; (B) the same coordinates grouped by source. Values are identical across the three seeds.

D.6 How the Response Profiles Are Drawn

Figure 3 (Section 4.2) draws one radar per model and a consensus panel. Each vertex is a protocol's primary response divided by its local clean lift L_0 (three-seed means, capped at 1.25), so the dashed ring marks a loss equal to the full lift. S3.4 has no lift; its vertex is the normalized entropy divided by 0.40. The figure is an index into the evidence: the readings in each protocol's own units are in the task scorecards (Appendix C.1) and in Sections D.1–D.5.

E Calibration on Planted Ground Truth

Main-text anchors: Section 4.1 reports that three protocols recover a planted answer and fail as predicted under faults; this appendix gives the planted tasks, the fault conditions, and the full readings. Section 3.1 places calibration in the pipeline.

Real EEG gives no ground truth for which channels, band, or window carry a label, so a reading on the campaign data cannot be checked against a known answer. The two stages are therefore also run on synthetic EEG whose label mechanism is known by construction: a 21-channel 10–20 montage, $256\,\text{Hz}$, $10\,\text{s}$ windows, a $1/f$ background with slow, theta, and beta rhythms at about $10\,\mu\text{V}$ per channel, and a balanced binary label carried by the amplitude ratio of two sinusoidal carriers that are present in both classes with matched total power, so that removing the carrier leaves the two classes identically distributed and total energy carries nothing (Table E.1). The frozen backbone and affine readout are fitted and locked as in Stage I, a route is read on a protocol's family of tasks where it reaches 0.75 on at least one of them, and every intervention passes through the Stage II operators of Appendix B unchanged. Beside the correct route, three fault conditions each break one link and state in advance what the diagnostic must then report (Table E.2); a protocol passes where it recovers the planted structure on the correct route and reports the predicted failure under the fault. No reading from these runs enters the campaign.

Table E.1: **Planted tasks.** Every task is a balanced binary discrimination read by AUROC (chance 0.5).

Task	Where the label is planted (carriers)	Read by	Predicted reading on the correct route
Spatial, frontal	The seven frontal channels FP1, FP2, F7, F3, FZ, F4, F8 ($21/23$ Hz); every other channel carries the pair at a label-independent ratio	S1.1	Collapse to chance at the full frontal mask; no change under the equal-channel scattered control; peak at frontal
Spatial, parietal	The five parietal channels P7, P3, PZ, P4, P8 ($21/23$ Hz)	S1.1	As above, with the peak at parietal
Spatial, shortcut	The frontal plant plus a label-carrying cue on one occipital channel, O2 ($21/23$ Hz)	S1.1	A readout wired to the plant peaks at frontal and one wired to the cue peaks at occipital: the diagnostic reads what a predictor uses
Temporal burst	One energy-matched burst per window, late in the window, on all channels; the class is the burst's carrier ($14/24$ Hz)	S2.1	Chance when the aperture excludes the burst; the response tracks event coverage, not duration; repeat fill can duplicate the burst
Spectral alpha	All channels ($9.5/11.5$ Hz)	S2.3	The alpha band-stop attenuates the carriers without erasing them; band-stops outside alpha leave the response unchanged

Table E.2: **Fault conditions run beside the correct route.**

Condition	What is broken	Reaches	Predicted reading
F1 shuffled channel map	The link between electrode identity and array row: the operator receives a permuted channel map while the readout keeps its wiring	S1.1, S1.2	The recovered peak leaves the planted region for the region the permutation sends the planted rows to
F3 wrong sampling rate	The link between declared and physical time: 256 Hz data declared as 200 Hz	S2.1, S2.3	Apertures span $0.78\times$ the intended duration and cutoffs land at $1.28\times$ the intended frequency; a diagnostic that reports the same band boundaries under both rates reads condition labels, not physical frequencies
F4 label independent	The link between signal and label: every component is generated as on the correct route and the label is drawn independently	All	Chance at every condition, including clean; the one chance anchor that removes the label by construction rather than by attenuation

E.1 COVERAGE

Table E.3 lists the validation-split clean synthetic AUROC (Gate G1) of every route over three Stage I seeds; the routes read in Section 4.1 follow from the rule above, and the label-independent condition (F4) sits at chance on every task.

Table E.3: **Gate G1 validation AUROC per route** (validation split, mean ± sample standard deviation over three seeds). Routes read on the spatial task: CBraMod, CSBrain, REVE; on the spectral task: BIOT, CSBrain, LaBraM, REVE; on the temporal task: all five. The 75 F4 cells average 0.485 ± 0.025 (maximum 0.550).

Task	BIOT	CBraMod	CSBrain	LaBraM	REVE
Spatial, frontal	0.610 ± 0.006	0.830 ± 0.000	0.766 ± 0.002	0.718 ± 0.017	0.773 ± 0.008
Spatial, parietal	0.568 ± 0.007	0.712 ± 0.002	0.666 ± 0.007	0.635 ± 0.019	0.666 ± 0.009
Spatial, shortcut	0.711 ± 0.017	0.873 ± 0.002	0.795 ± 0.003	0.735 ± 0.018	0.790 ± 0.013
Temporal burst	1.000 ± 0.000	0.899 ± 0.001	1.000 ± 0.000	0.998 ± 0.003	0.997 ± 0.003
Spectral alpha	0.997 ± 0.000	0.659 ± 0.004	1.000 ± 0.000	0.990 ± 0.007	0.971 ± 0.006
F4, all tasks	0.482 ± 0.031	0.502 ± 0.028	0.485 ± 0.030	0.472 ± 0.011	0.482 ± 0.013

E.2 Spatial Calibration

Figure E.1 and Table E.4 give the spatial calibration for the three routes read on the planted spatial tasks. Across 27 model–task–seed cells, complete masking of the planted region gives AUROC 0.498–0.581 and the equal-channel scattered control 0.747–0.883, against clean 0.713–0.862. Replacing zeroing by a power-conserving temporal shuffle of the masked channels gives SCS $+0.211$ to $+0.332$ against $+0.216$ to $+0.349$ under zeroing. In the frontal-by-parietal matrix (Figure E.1C), the two diagonal cells are $+0.308 \pm 0.033$ and $+0.240 \pm 0.023$ and the eight off-diagonal cells lie in -0.096 to $+0.024$; the negative off-diagonal values scale with how many planted channels the scattered draw happened to hit (0, 1, or 2 channels: $+0.002$, -0.026, -0.076). Under the permuted channel map (F1), SCS is -0.089 to -0.177 in every diagonal cell, the recovered peak moves to the region the permutation sends the planted rows to in 27 of 27 cells, and the task itself stays learnable (frontal clean AUROC 0.753–0.843). On the shortcut task, masking the seven frontal channels gives SCS $+0.255$ to $+0.321$, while masking the occipital pair that carries the cue gives $+0.004 \pm 0.018$; the operational atlas cannot mask O2 alone, which bounds this reading.

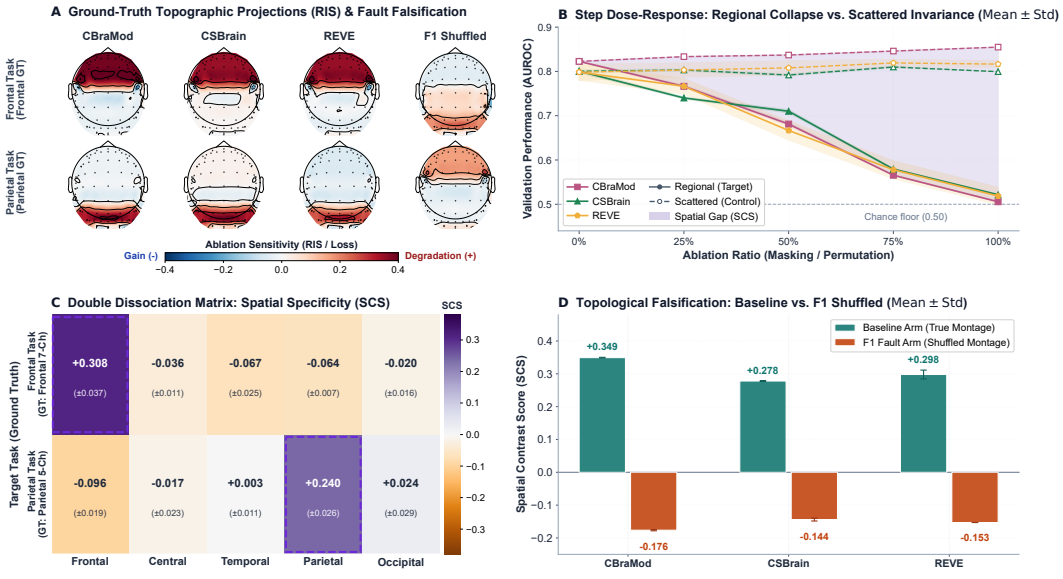


Figure E.1: **Spatial calibration on planted regions** (three seeds, mean ± sd). (A) Regionwise RIS on the frontal and parietal tasks for the three routes read, and under the permuted channel map (F1). (B) AUROC against masking ratio: planted region (solid) against the equal-channel scattered control (dashed); the shaded gap is SCS. (C) SCS for the frontal and parietal tasks against the five regions. (D) SCS at the full mask under the correct channel map and under F1.

Table E.4: **Spatial calibration on the test split** (three seeds, mean ± sd). Masked columns are at the full mask; the F1 column for REVE is a single seed.

		Masked (AUROC)		SCS (AUROC points)		
Task	**Clean**	**Planted region**	**Scattered**	**Zero**	**Shuffle**	**Under F1**
CBraMod						
frontal	0.823±0.001	0.506±0.001	0.855±0.000	+0.349±0.001	+0.332±0.003	−0.176±0.002
parietal	0.767±0.001	0.525±0.003	0.793±0.003	+0.268±0.004	+0.261±0.003	−0.153±0.001
shortcut	0.862±0.002	0.561±0.003	0.882±0.001	+0.321±0.003	+0.317±0.003	−0.148±0.001
CSBrain						
frontal	0.800±0.003	0.522±0.001	0.799±0.000	+0.278±0.002	+0.249±0.007	−0.144±0.005
parietal	0.770±0.002	0.527±0.001	0.764±0.001	+0.237±0.001	+0.211±0.005	−0.104±0.003
shortcut	0.810±0.002	0.539±0.001	0.819±0.005	+0.280±0.005	+0.265±0.002	−0.153±0.003
REVE						
frontal	0.798±0.018	0.519±0.020	0.816±0.008	+0.298±0.015	+0.230±0.008	−0.153
parietal	0.713±0.007	0.536±0.003	0.751±0.007	+0.216±0.009	+0.227±0.012	−0.089
shortcut	0.822±0.003	0.570±0.010	0.825±0.010	+0.255±0.020	+0.281±0.019	−0.123

E.3 Temporal Calibration

Figure E.2 and Table E.5 give the temporal calibration for all five routes. Centered apertures give AUROC 0.499–0.528 at $2\,\mathrm{s}$ and 0.508–0.520 at $4\,\mathrm{s}$, where the window holds 0 and 0.6% of the burst energy, and 0.882–1.000 at $6\,\mathrm{s}$, where it holds 99.4%. Random apertures follow the expected coverage (0.25, 0.42, 0.62, 1.0 for 2, 4, 6, $8\,\mathrm{s}$), and a random $2\,\mathrm{s}$ aperture (0.60–0.77) scores above a centered $4\,\mathrm{s}$ one (0.51–0.52). Random placement raises the burst reading by 0.07–0.26 at $2\,\mathrm{s}$ and 0.18–0.32 at $4\,\mathrm{s}$ across the five routes, and changes the four stationary tasks by -0.14 to $+0.02$; fill policy moves only the stationary tasks, where zero fill costs 0.00–0.24 AUROC points at $2\,\mathrm{s}$, while on the burst the two fills differ by at most 0.025 at $2\,\mathrm{s}$ and 0.033 anywhere on the grid. Under the mis-declared sampling rate (F3), the declared $6\,\mathrm{s}$ aperture spans $4.69\,\mathrm{s}$ and 23.8% of the burst energy, which the operator audit records (visible fraction $0.600 \to 0.469$); the $6\,\mathrm{s}$ reading then moves by 0.000 (BIOT), 0.012 (LaBraM), 0.025 (CSBrain), 0.095 (REVE), and 0.211 (CBraMod). Within this range the aperture reading follows how much of a transient event the window holds; on stationary content the zero-fill curve carries a fill cost and is read beside the repeat-fill curve.

Table E.5: **Temporal calibration on a planted burst** (three seeds, mean ± sd). AUROC of centered apertures under zero fill. Under F3 the declared $6\,\mathrm{s}$ window covers $[2.66, 7.34]\,\mathrm{s}$.

Aperture	**Burst energy**	**BIOT**	**CBraMod**	**CSBrain**	**LaBraM**	**REVE**
Full ($10\,\mathrm{s}$)	1.000	1.000 ± 0.000	0.870 ± 0.001	1.000 ± 0.000	0.997 ± 0.004	0.994 ± 0.005
$2\,\mathrm{s}$	0	0.528 ± 0.001	0.499 ± 0.002	0.508 ± 0.009	0.503 ± 0.020	0.503 ± 0.007
$4\,\mathrm{s}$	0.006	0.519 ± 0.023	0.509 ± 0.001	0.520 ± 0.002	0.508 ± 0.016	0.520 ± 0.014
$6\,\mathrm{s}$	0.994	1.000 ± 0.000	0.882 ± 0.002	1.000 ± 0.000	1.000 ± 0.001	0.965 ± 0.017
$8\,\mathrm{s}$	1.000	1.000 ± 0.000	0.875 ± 0.001	1.000 ± 0.000	0.999 ± 0.002	0.986 ± 0.009
$6\,\mathrm{s}$ under F3	0.238	1.000 ± 0.000	0.671 ± 0.002	0.975 ± 0.001	0.987 ± 0.007	0.870 ± 0.023
Shift at $6\,\mathrm{s}$	–	0.000 ± 0.000	0.211 ± 0.004	0.025 ± 0.001	0.012 ± 0.008	0.095 ± 0.038

E.4 Spectral Calibration

Figure E.3 and Table E.6 give the spectral calibration for the four routes read on the planted alpha task. The band-stop attenuates the carriers to $-30\,\mathrm{dB}$ rather than removing them; an ideal receiver still scores 0.979 under the alpha band-stop, so a collapse measures the route's sensitivity to the residual, not the absence of the band. The alpha band-stop takes the four routes to AUROC 0.499–0.528 (collapse 0.36–0.50), and the four bands without a carrier stay within -0.019 to $+0.027$. The one-sided sigma band-stop ($11.5\,\mathrm{Hz}$ carrier attenuated, $9.5\,\mathrm{Hz}$ intact) gives a collapse of 0.35–0.48 for BIOT, LaBraM, and REVE and 0.108 for CSBrain, while the $8\,\mathrm{Hz}$ low-pass orders the routes differently (retention 0.82, 0.66, 0.75, 0.51 for BIOT, CSBrain, LaBraM, REVE), so the two conditions read different properties. Under the mis-declared sampling rate (F3) the theta column rises

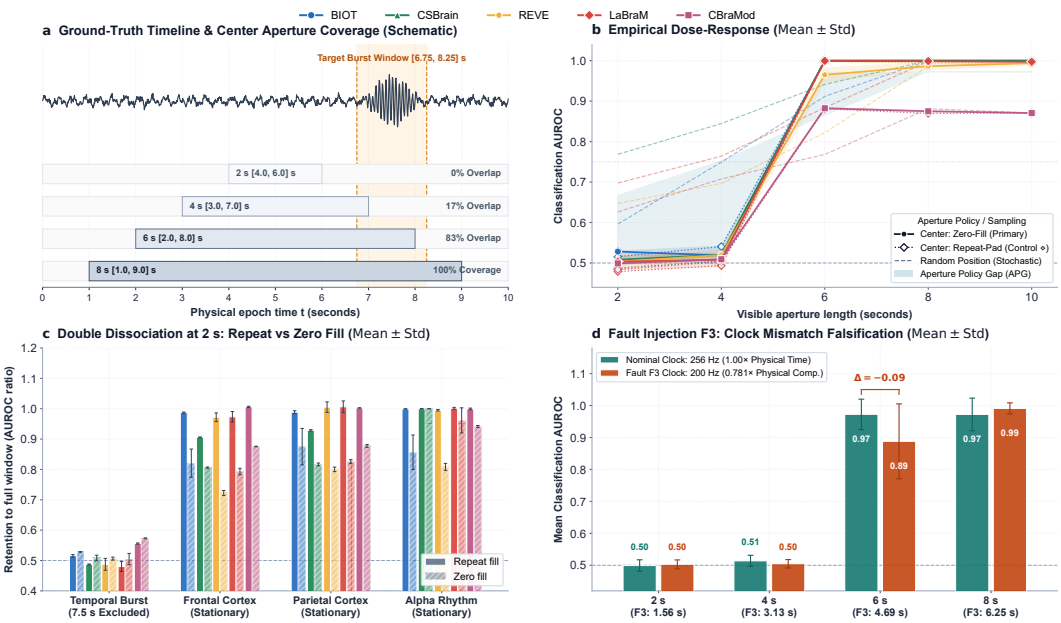


Figure E.2: **Temporal calibration on a planted burst** (three seeds, mean ± sd). (A) Burst timeline and centered apertures. (B) AUROC against aperture length under zero fill, repeat fill, and random placement. (C) Retention at 2 s on the burst task and on three stationary tasks, repeat fill and zero fill side by side. (D) The centered readings under the correct and the mis-declared sampling rate, averaged over the five routes; per-route values are in Table E.5.

to 0.15–0.38 and the sigma column falls to within 0.021 of zero, so the collapse follows the physical carrier frequency. The full-pass reference of S2.3 is the RMS-normalized input; that normalization costs BIOT and REVE nothing, LaBraM 0.15–0.21 on the alpha task, and CSBrain 0.22–0.26 on the spatial tasks.



Figure E.3: **Spectral calibration on a planted alpha pair** (three seeds, mean ± sd). (A) Retention against low-pass cutoff. (B) Band-stop collapse by nominal band. (C) Theta collapse under the correct and the mis-declared sampling rate. (D) Band-stop collapse under the mis-declared rate, bands in physical hertz.

Table E.6: **Spectral calibration on a planted alpha pair** (three seeds, mean ± sd). Collapse = 1 − retention under each band-stop. Cells without a spread have a sample standard deviation below 0.001.

	BIOT	**CSBrain**	**LaBraM**	**REVE**
Full AUROC	0.998 ± 0.000	0.997 ± 0.001	0.780 ± 0.042	0.977 ± 0.004
Collapse under band-stop				
δ	$+0.001$	-0.002	-0.019 ± 0.006	$+0.027 \pm 0.007$
θ	-0.000	-0.000	-0.002	$+0.020 \pm 0.001$
α (carrier)	$+0.471 \pm 0.004$	$+0.500 \pm 0.007$	$+0.357 \pm 0.027$	$+0.474 \pm 0.012$
σ	$+0.377 \pm 0.068$	$+0.108 \pm 0.007$	$+0.350 \pm 0.039$	$+0.480 \pm 0.038$
β	$+0.004$	-0.000	-0.008 ± 0.007	$+0.004$
γ	$+0.002$	-0.000	-0.000	$+0.001$
Collapse under F3, bands in physical hertz				
θ	$+0.152 \pm 0.078$	$+0.180 \pm 0.008$	$+0.374 \pm 0.079$	$+0.251 \pm 0.047$
σ	$+0.000$	-0.014	-0.021	$+0.001$

E.5 Scope and Reading

AUROC, SCS, and aperture-policy gaps are in AUROC points; RIS, retention, and collapse are ratios. Three readings follow from the calibration. A negative SCS is read as the absence of regional specificity, not as a regional response; this is the reading the F1 condition fixes. On stationary content the zero-fill aperture curve is a lower bound on temporal dependence and is read beside the repeat-fill curve. A band-stop collapse is the route's sensitivity to the attenuated residual and is reported with the residual level, not as the removal of a band. The calibration covers one operator seed and one scattered draw per configuration, one event geometry on the temporal task, one carrier pair on the spectral task, retrained readouts under F3, and no test of transfer from synthetic to recorded EEG.

F From Diagnosis to Model Design: Readout Refits

Main-text anchor: Section 5.1 asks whether a diagnosed dependence can be repaired at the readout while the backbone stays frozen (Figure 7); this appendix gives the two refits behind that answer, S1.2 on Motor and S3.2 on Seizure. The campaign readings they start from are in Section 4.3 and Appendix D.1.

Scope and locking boundary. Stage I's selection rule governs how a state is produced. The refits stay outside the campaign: no reading below enters the campaign or changes a figure in Section 4. It continues an already-selected readout for a declared, equal step budget and reads the result against a control state that receives the same budget. Validation-selected stopping is not used, because per-state stopping would let the two states differ in optimization steps and the target-minus-control contrast would no longer isolate the intervention. The states below stay inside the frozen-backbone, affine-readout family of Appendix A.7.

F.1 Readout Symmetry Refit (S1.2)

Design. Section 4.3 reports that hemispheric exchange drives CBraMod's and CSBrain's Motor AUROC to 0.471 and 0.480 while reversing $44.3\% \pm 1.1\%$ and $39.8\% \pm 6.3\%$ of their left/right predictions. Two explanations remain open at that point: the frozen backbone may not preserve the left–right geometry the reflection acts on, or the readout may be fitted without the constraint that would use it. Three readout states separate them on one frozen backbone. The *shipped* state is the Stage I readout, unchanged. The *control* state continues that readout on batches doubled without reflection. The *mirror* state continues it on batches doubled with the same left–right montage reflection the diagnosis applies (Appendix B.9): every complete homologous pair the montage carries is exchanged across the sagittal midline (C3 $\leftrightarrow$ C4, FC3 $\leftrightarrow$ FC4, and the remaining lateral pairs), midline and unmatched channels stay in place, and the paired left$\leftrightarrow$right label permutation follows that exchange, with both-fist and foot labels fixed. Control and mirror share optimizer, schedule, and step budget, so the mirror-minus-control contrast isolates the reflection rather than the fact of continued training. Evaluation uses the fixed `physiomi natural_test` cohort ($1{,}893$ trials, of which 945 carry a left or right label) and the three Stage I seeds.

Enforced invariants. The loader rejects any parent checkpoint whose selection bank does not read `natural_valid` identically in the run configuration, the best-checkpoint metadata, and the checkpoint extra, so every state descends from a validation-selected Stage I state. Each training epoch emits a per-task gradient audit, and the run aborts if that audit is missing or if any backbone gradient norm exceeds zero, so the frozen backbone is a recorded measurement rather than a claim in prose. The shipped state reproduces the campaign readings exactly (clean AUROC 0.750 ± 0.002 and 0.748 ± 0.008; reversal rates 0.443 ± 0.011 and 0.398 ± 0.063), so the refit starts from the state the main text reports.

Reversal has a state-specific ceiling. A reversal requires the original prediction to fall on a lateral class, so the reversal rate is bounded above by the share of true left/right trials that the state already assigns to left or right. That ceiling is a property of the state, not a constant: across the four readable shipped states it runs from 0.60 to 0.72, and it is computed per run from the exported predictions rather than assumed to be 1. Realization, the ratio of the reversal rate to its ceiling, is therefore the quantity that reports how much of the achievable symmetry a state actually uses.

The gain is in realization, not in the ceiling. A higher reversal rate could follow from a state that simply assigns more trials to a lateral class. Decomposing each isolating contrast into a ceiling term, a realization term, and a cross term separates the two (Table F.1). The realization term carries $+0.157$, $+0.108$, $+0.110$, and $+0.077$ of totals of $+0.199$, $+0.094$, $+0.140$, and $+0.091$ for CBraMod, CSBrain, LaBraM, and REVE; for CSBrain it exceeds the total, because that state's ceiling falls by 0.011. Across the four models the mirror refit raises the share of achievable reversals from 0.467–0.688 to 0.719–0.868 while the ceiling itself moves by at most 0.048. A single affine layer over an unchanged backbone is therefore enough to reach the left–right structure, which places the shipped failure at the readout rather than at a geometry the backbone did not retain.

Table F.1: **Readout symmetry refit on Motor Imagery** (S1.2; three Stage I seeds, mean ± sd). PRR and realization (PRR divided by the state's reversal ceiling) are read under the hemispheric swap; clean AUROC is read on the unswapped montage.

	PRR			Mirror − control		Realization
Model	shipped	control	mirror	ΔPRR	ΔClean	shipped → mirror
CBraMod	0.443 ± 0.011	0.365 ± 0.019	0.564 ± 0.023	$+0.199$	$+0.030$	$0.688 \rightarrow 0.868$
CSBrain	0.398 ± 0.063	0.411 ± 0.020	0.505 ± 0.028	$+0.094$	$+0.016$	$0.658 \rightarrow 0.801$
LaBraM	0.291 ± 0.069	0.313 ± 0.041	0.453 ± 0.011	$+0.140$	$+0.020$	$0.467 \rightarrow 0.719$
REVE	0.383 ± 0.122	0.488 ± 0.013	0.579 ± 0.013	$+0.091$	$+0.017$	$0.537 \rightarrow 0.798$
BIOT[†]	0.035 ± 0.009	0.034 ± 0.023	0.031 ± 0.023	-0.002	$+0.005$	$0.087 \rightarrow 0.080$

[†] BIOT has no lift on this task (clean AUROC 0.511 ± 0.008, $L_0 = 0.011$); its row is reported as measured and not read.

What the control absorbs. The control state is not inert, and reading against the shipped readout instead would misreport every model. Continued training alone moves the reversal rate by -0.078 for CBraMod and $+0.105$ for REVE, and moves the clean score by -0.026 and $+0.011$ respectively. Measured against the shipped readout the mirror state's clean AUROC gain is $+0.004$ to $+0.036$ rather than the $+0.016$ to $+0.030$ that the equal-budget contrast reports. Table F.1 carries the pre-specified contrast; the shipped-relative reading is stated here so that neither is mistaken for the other.

What the states establish. The contralateral reflection prior is reachable through the affine readout of a frozen backbone on `physiomi natural_test`, and the equal-budget control separates that recovery from continued readout training. The backbone is bit-identical across the three states, so the result places the shipped failure at the readout; the clean-score movement is the regularization a paired augmentation supplies to a small readout, and the result does not transfer to a task that carries no lateralization prior.

F.2 Readout Robustness Refit (S3.2)

Design. Appendix D.4 reports that injected ocular and myogenic waveforms move Seizure scores by model- and source-specific amounts. The same two explanations are open: the frozen backbone may not separate the artifact from the signal, or the readout may never have been asked to. Four readout states are fitted on one frozen backbone under the equal-budget rule above. The *shipped* state is the Stage I readout. The *control* state continues it on batches doubled without contamination. The *artifact* state continues it on batches doubled with both injected sources, and a fourth *EMG-only* state repeats that with myogenic contamination alone in order to test transfer across artifact type. Contamination during refitting uses signal-to-artifact ratios $\{5, 2\}$ only; the S3.2 grid is evaluated at $\{10, 5, 2, 1, 0.5\}$, so $\{10, 1, 0.5\}$ are held out at both ends of the severity range. Labels are untouched, so the property under test is invariance rather than equivariance. Evaluation uses the fixed `siena natural_test` cohort and the three Stage I seeds.

Reading this section against the campaign. Appendix D.4 reports S3.2 through $\mathrm{AUC}_a^{\log}$, the area under the log-dose curve of primary-metric loss. This section reports retention, $1 - \mathrm{gap}/(Q_0 - \mathrm{floor})$, evaluated at each dose and averaged over the grid. The two quantities are not interchangeable: retention is normalized by baseline lift and read dose by dose, so a state with a small lift can move far in retention at a small $\mathrm{AUC}_a^{\log}$, and retention above 1 records a contaminated score above its own clean reference rather than a gain in capability. The two quantities order the shipped states differently: ocular $\mathrm{AUC}_a^{\log}$ is largest for BIOT, then REVE, then LaBraM, whereas ocular retention below is lowest for LaBraM (0.727), then BIOT (0.772), then REVE (0.865).

Recovery reaches the doses it was not trained on. Where a state moves at all, the held-out doses move with the trained ones. LaBraM's ocular retention rises by 0.286 over the full grid and by 0.272 over the three withheld doses; REVE's rises by 0.047 and 0.060; BIOT's myogenic retention rises by 0.070 on both. Refitting at two intermediate doses therefore does not leave a state tuned to those

Table F.2: **Readout robustness refit on Seizure Detection** (S3.2; three Stage I seeds, mean ± sd). Retention is $1 - \text{gap}/(Q_0 - \text{floor})$ averaged over the dose grid. ΔClean and the refit − control columns are contrasts against the equal-budget control; held-out averages the three doses withheld from refitting.

		Myogenic (EMG) retention			Ocular (EOG) retention		
			Refit − control			Refit − control	
Model	**ΔClean**	Shipped	all doses	held-out	Shipped	all doses	held-out
CBraMod	−0.002	1.034 ± 0.013	−0.008	−0.008	0.998 ± 0.007	−0.015	−0.018
CSBrain	−0.064	0.904 ± 0.068	−0.018	−0.009	0.939 ± 0.033	−0.093	−0.131
LaBraM	+0.014	1.018 ± 0.030	−0.069	−0.027	0.727 ± 0.053	+0.286	+0.272
REVE	−0.066	0.981 ± 0.019	−0.035	−0.037	0.865 ± 0.038	+0.047	+0.060
BIOT	+0.002	0.864 ± 0.013	+0.070	+0.070	0.772 ± 0.114	−0.014	−0.004

doses: the gain is present at held-out doses on both sides of the training grid. Panels A and B of Figure F.1 show the curves, with trained doses shaded and held-out doses drawn as open markers.

Recovery is specific to the artifact it was refitted on. No state improves on both sources. LaBraM gains 0.286 on ocular contamination while losing 0.069 on myogenic; BIOT gains 0.070 on myogenic while losing 0.014 on ocular. Refitting on myogenic contamination alone and reading the result on ocular contamination separates the two further: that transfer costs LaBraM 0.250 and REVE 0.069, leaves BIOT unchanged (−0.005), and gives CBraMod and CSBrain +0.012 and +0.037. What the readout acquires is a projection away from the waveform it was shown, not a general inattention to non-neural channels, and a deployment whose contamination differs from the refitting source cannot assume the gain carries.

Refitting is not free, and for two states it is negative. The clean score pays for the augmentation in REVE (−0.066) and CSBrain (−0.064), and CSBrain's ocular retention also falls, by 0.093 over the grid and 0.131 over the held-out doses, so that state is worse on both axes than its equal-budget control. CBraMod enters at ocular and myogenic retention of 0.998 and 1.034 and has nothing to recover; its movements are within the seed spread. Read as a decision, the five states divide into one that gains on both axes for its refitted source (LaBraM, and BIOT for myogenic), one already invariant (CBraMod), one that buys robustness with clean score (REVE), and one that loses on both (CSBrain). Panel C of Figure F.1 places them on those two axes, using each state's larger-magnitude source; the per-source values are in Table F.2.

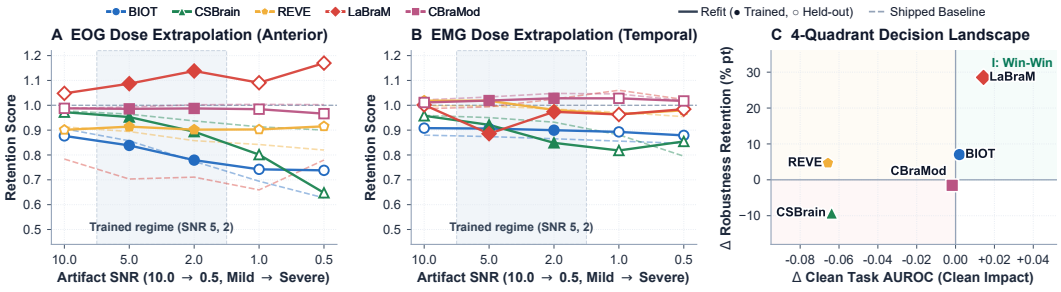


Figure F.1: **Readout robustness refit** (S3.2, Seizure detection). (A, B) Retention against signal-to-artifact ratio for ocular and myogenic contamination. Solid lines are the artifact-refitted state and dotted lines the shipped readout; the shaded band and filled markers mark the two doses used in refitting, and open markers the three held-out doses. (C) Change in retention against change in clean AUROC, both relative to the equal-budget control, using each state's larger-magnitude source: myogenic for BIOT, ocular for the other four. Per-source values are in Table F.2.

What the states establish. Robustness to the injected procedural artifact is reachable through the affine readout of a frozen backbone on `siena natural_test` for some states and not for others,

and the equal-budget control separates that recovery from continued readout training. The backbone is bit-identical across the four states, and the artifacts are the synthetic waveforms of S3.2 injected into anatomically selected channels, not recorded ocular or myogenic activity; nothing here re-ranks the campaign.

G Comparison with Concurrent EEG-FM Evaluations and Audits

Main-text anchor. Section 1 identifies three obstacles in evaluating EEG foundation models: route confounding, lack of ground truth, and lack of attribution. Section 2 grounds our protocols in three physiological properties: spatial topography, rhythm structure, and artifact dynamics. This appendix places our design beside concurrent evaluations along these axes.

Scope note. Every study compared here is concurrent with this work: each appeared as a preprint in 2026 (Širca et al., 2026; Kontras et al., 2026; Wang et al., 2026; Tang et al., 2026; Lin et al., 2026; Yu et al., 2026; Zare, 2026; Lehn-Schiøler et al., 2026). They are evaluated as parallel explorations of model behavior, not as prior art.

How the table is read. Table G.1 summarizes eight concurrent studies and this work across six diagnostic capabilities (columns 2–7) and two boundaries (columns 8–9). The marks reflect the questions each study chose to address, not a quality ranking.

Confounding: Decoupled stages, input interface, and matched controls. Downstream evaluation routes introduce confounding. Replacing the readout changes model scores and can reverse rankings across benchmarks on the same data (Xiong et al., 2026; Lahiri et al., 2026).

Our framework addresses route confounding through a two-stage decoupled design. Stage I locks one state per route (adapter, frozen backbone, and linear readout) under shared rules. Stage II evaluates that fixed state across eight protocols without training. In contrast, existing benchmarks either evaluate multiple regimes across tasks (Širca et al., 2026) or train models downstream per task (Wang et al., 2026), whereas our eight protocols run across a single canonical state held strictly fixed throughout.

Physical interventions also introduce confounding side effects. Masking electrodes removes target brain regions, but also reduces channel count; truncating a window introduces boundary padding; adding background signals increases energy. Prior perturbation setups (Širca et al., 2026) do not isolate these operation-specific side effects from the targeted physiological dependencies. In our design, every operation with a side effect is paired with a pre-registered control: regional masking is paired with an equal-channel scattered control, zero-fill is paired with repeat-fill, and background mixing is matched in RMS amplitude. On our zero-masking grid, 32.0% of the 206 regional maskings that cost more than 0.05 metric points are met or exceeded by their equal-channel control (Appendix C.2), showing that a regional drop often reflects channel loss rather than regional specificity. Where an operation introduces no separable side effect (e.g., swapping homologous channels in S1.2), the protocol specifies a reference alone (Table B.4); where no matched condition can isolate the side effect, the limit is stated as applicability (Appendix B.4).

All interventions act on the $C \times T$ signal before resampling and input representation. This allows the same physical change to reach all five models. In contrast, internal probes (Tang et al., 2026; Lin et al., 2026; Lehn-Schiøler et al., 2026) depend on each model's specific hidden dimensions and sequence layouts.

Ground truth: Planted calibration versus random nulls. Real EEG provides no physiological ground truth for which channel, frequency band, or time window carries a label.

Concurrent audits address this by defining null baselines: shuffled targets (Tang et al., 2026), random subject variance (Lin et al., 2026), or permuted labels (Zare, 2026). A null baseline establishes what a reading must not return. However, it cannot verify whether a protocol recovers the true dependency when one is present.

Our pipeline is calibrated on synthetic EEG with planted ground truth. The discriminative signal is planted in a declared set of channels, frequency band, or time window, alongside three fault

conditions: permuted channels, mis-declared sampling rates, and independent labels. The protocols recovered the planted dependencies and failed under the predicted fault conditions (Section 4.1). This establishes what the readings mean before evaluating recorded EEG.

Attribution: Two-layer closed-loop action. A diagnostic drop alone does not indicate how to improve a model. Without controlling intermediate states, one cannot tell whether a failure stems from the interface or the backbone.

We close the loop at two layers. At the interface, refitting only the readout on mirrored batches raised hemispheric-reversal realization from 0.47–0.69 to 0.72–0.87 against an equal-budget control on the unchanged backbone (Appendix F.1), and adapter normalization determines whether the input amplitude reaches the backbone. Both provide immediate deployment guidelines on the existing model without retraining the backbone. When a failure is rooted in the backbone, we specify a named pretraining target with a test that could reject it (Section 5).

Among concurrent works, FAME (Yu et al., 2026) closes the loop by correcting low-frequency bias in pretraining, and The Identity Trap (Lin et al., 2026) erases the subject identity subspace. We extend closed-loop action to distinguishing interface-level fixes (S1.2, S3.2) from pretraining targets across architectures.

Profile versus ranks, and explicit prices. Leaderboards rank models by scalar scores. However, scalar scores conceal shared behavioral preferences and fail to differentiate models with near-identical scores (Section 4.3). Our benchmark outputs a multi-dimensional response profile across three physical dimensions, reporting each reading beside its reference, its control where the protocol has one, and its baseline lift L_0.

This design incurs explicit trade-offs:

1. **Input versus internal access:** Acting at the input enables uniform evaluation across architectures, but forecloses internal attribution and feature dictionary extraction (Širca et al., 2026; Tang et al., 2026; Lehn-Schiøler et al., 2026). Protocol S3.4 is our only reading of representation geometry.
2. **Single versus multiple regimes:** Holding one canonical state fixed isolates the properties of the fixed route, but does not measure how full fine-tuning modifies these dependencies (Širca et al., 2026; Lin et al., 2026).
3. **Diagnostic depth versus cohort breadth:** We evaluate five models across four representative tasks in depth, rather than sweeping dozens of datasets with scalar scores (Kontras et al., 2026); each task uses one source cohort, and cross-cohort variations remain for future investigation (Zare, 2026).

Table G.1: **Comparison with concurrent EEG-FM evaluations and audits.** Columns 2–7 represent the core diagnostic capabilities of this benchmark; columns 8–9 record our deliberate trade-offs and boundaries. ● carried as a design element under the study's own question; ○ carried in part, on one axis, or beside another regime; – not part of the question the study set itself. The marks reflect each study's declared question, not a quality ranking.

Study	Physiological Dimensions	Decoupled State	Input Interface	Matched Controls	Planted Calibration	Closed-Loop Action	Direct Latent Access	Scale and Focus
NeuroAtlas (Kontras et al., 2026)	– *clinical and BCI task transfer*	● *frozen backbone, linear probe per fold*	– *clean evaluation only; no perturbation*	– *no perturbation*	– *held-out validation folds*	– *reports transfer gap; datasets released*	–	42 datasets, ∼260k hours; transfer leaderboard
Beyond Accuracy (Širca et al., 2026)	○ *spatial dropout and additive noise*	● *head-only on frozen backbone, and full fine-tuning*	○ *input perturbations; white-box AttnLRP*	○ *control region; no equal-channel control*	○ *qualitative comparison with known biology*	○ *pooling named as cause of weak linear score*	● *AttnLRP; block-wise probes*	6 EEG-FMs, 8 datasets; robustness and interpretability
EEG-FM-Audit (Wang et al., 2026)	○ *temporal, spatial, spectral; no background*	○ *models fine-tuned downstream per task*	● *surrogates: phase, ROI noise, band cut*	○ *phase-randomized surrogate preserves covariance and spectrum*	–	○ *ablation of pretraining paradigms*	–	4 EEG-FMs, 5 supervised baselines, 3 datasets
What Do EEG-FMs Capture? (Tang et al., 2026)	○ *six clinical feature families*	● *frozen backbone, linear head*	– *internal ridge probes and LEACE erasure*	○ *shuffled- and Gaussian-target nulls*	– *random nulls only; no planted ground truth*	○ *unencoded residual named for discovery*	● *layer-wise probes; concept erasure*	3 EEG-FMs, 5 clinical tasks, 945 triplets
The Identity Trap (Lin et al., 2026)	○ *subject identity; aperiodic $1/f$ slope*	● *frozen backbone; fine-tuned state also read*	○ *embedding probes and aperiodic ablation*	○ *random null for subject variance*	– *random null for subject variance*	● *subject axis erased, decoding re-read*	● *variance decomposition; subspace erasure*	3 EEG-FMs, 4 datasets in 2×2 design; identity audit
Negative-Control (Zare, 2026)	– *dataset identity, cohort, montage*	● *frozen encoders*	– *subject-disjoint embedding probes*	○ *permuted labels; random encoder*	– *negative controls for dataset transfer*	○ *reporting protocol distilled*	○ *dataset identity decoded from embeddings*	5 encoders, 5 tasks, 1 external cohort (1,187 recordings)
SAE Dictionaries (Lehn-Schiøler et al., 2026)	– *clinical concepts; latent edits as spectra*	● *frozen embeddings; concept probes*	– *sparse autoencoders on embeddings*	–	– *intrinsic dictionary audit*	○ *concept steering exposes entanglement*	● *TopK sparse autoencoders; steering*	3 EEG-FMs; 4 clinical concepts; mechanistic interpretability
FAME (Yu et al., 2026)	○ *spectral bands*	● *linear probing of pretrained encoders*	– *embedding frequency preference*	–	– *downstream score after correction*	● *corrected in pretraining; re-evaluated*	○ *embedding frequency preference*	41 downstream tasks; low-frequency bias correction
This work	● *space, temporal–spectral, background; 8 protocols*	● *locked in Stage I; Stage II trains nothing*	● *$C \times T$ before input transform; reaches all 5*	● *scattered masking, repeat fill, RMS match; reference where no side effect*	● *S1.1, S2.1, S2.3 on planted dependencies; 3 faults*	● *readout refit; pretraining targets with tests*	○ *S3.4 only: spectral entropy under non-brain input*	5 EEG-FMs, 4 tasks with one source each, 3 seeds, 8 protocols; one regime

H Reproducibility Protocol, Compute Footprint, and Release Integrity

Main-text anchor: Section 3.

H.1 Software and Hardware Environment

To support independent reruns and post-hoc audit, the benchmark records explicit entrypoints, seeds, manifests, completion checks, and the observed runtime family. These records identify what was fixed and what was not captured; they do not by themselves guarantee bit-identical execution on a different software or hardware stack. Table H.1 states that boundary.

Table H.1: **Observed software/runtime context and artifact-integrity scope.** Runtime ranges describe the archived campaign context; unrecorded details are not reconstructed as exact constraints.

Configuration Dimension	Preserved Specification / Audit Policy
Language & Runtime	Archived campaign code targets Python `3.11` and PyTorch `2.x`. Patch-level Python and PyTorch versions were not captured in a repository lockfile.
Scientific Ecosystem	Relevant paths use MNE-Python, SciPy, NumPy, and Scikit-learn. The current artifact does not claim a complete transitive dependency lock with package hashes; reruns must record their resolved environment beside generated outputs.
Compute Acceleration	Archived runs used Linux CUDA-compatible GPU nodes with one logical GPU per Stage I or Stage II worker. The benchmark suite employs linear probing over frozen representations, requiring modest compute resources suitable for standard workstation environments.
Seed Control	Stage I entrypoints receive an explicit training seed and call `torch.manual_seed(seed)`. Split/bootstrap seed `42` and Stage II runtime seed `20260513` are recorded separately where applicable; no blanket cross-library or cross-hardware bit-exactness is claimed.
Artifact Identity	Stage II campaign manifests bind allow-listed run directories to code identity, and completion receipts check coordinate coverage, status, and prediction inventory. SHA-256 digests are authoritative where recorded; a checkpoint without one is carried as an explicit missing value.

H.2 Deterministic Seed Hierarchy and Checkpoint Locking Invariants

The benchmark distinguishes between training-state variation (Stage I readout optimization) and perturbation-realization determinism (Stage II diagnostic interventions). Table H.2 outlines the random seed hierarchy, validation checkpoint locking criteria, and protocol binding rules.

Table H.2: **Random seed hierarchy and checkpoint locking rules.** Operational values, invariant rules, and deterministic bindings across Stage I and Stage II pipelines.

Protocol Dimension	Operational Value and Protocol Binding
Training Seeds ($N = 3$)	Three independent Stage I readout training seeds: `20260528` (visualization seed), `20260612`, and `20260718` (Section 3.3; Appendix B.7). Split construction and bootstrap procedures use seed `42` where specified; it is not a global training seed.
Perturbation Grids	Stage II physical parameter grids (mask ratios, cutoff frequencies, modulation factors, SNR doses) are deterministic. Stochastic realizations are protocol-seeded: channel selection uses `seeded_nested_random_v1`; S2.1 random window positions use runtime seed `20260515`; S3.2 synthetic artifact injection derives deterministic per-sample seeds from `runtime.seed=20260513` and `sample_id`.
Scattered Control Seeds	Scattered-control channel selections derive from the same `seeded_nested_random_v1` strategy using the per-condition runtime seed above; no ad-hoc manual seeds are introduced.
Checkpoint Locking Rule	Readout checkpoints are selected by highest mean balanced accuracy across the validation bank at epoch completion (25 epochs total); test partitions are strictly held out until frozen inference.

H.3 Execution Chain and Artifact Index

The release exposes wrapper-level entrypoints so that model-specific flags, frozen-backbone settings, run identifiers, and checkpoint directories are fixed in one place. The chain below runs from the repository root for the visualization seed. Stage II first preflights the five run directories and then emits an explicit campaign manifest. Final assembly refreshes condition-level evidence only from that allowlist and attains `final` status once the listed completion receipts pass.

```
# Stage I - five frozen backbones, training seed 20260528
bash scripts/model_training/canonical/run_5model_frozen_4source_v1.sh

# Stage II - five models x eight protocols
bash scripts/subbench/coordinated/run_5model_frozen_validsel_test_v1.sh

# Evidence assembly - refresh from that explicit campaign only
CAMPAIGN_DIR=outputs/subbench/campaigns
CAMPAIGN_ID=stage2_v007_4source_frozen_validsel_test_v1
python -m benchmark_core.evidence_profile.assemble_benchmark_evidence \
    --refresh-layer1 \
    --campaign-manifest "$CAMPAIGN_DIR/$CAMPAIGN_ID.json" \
    --output outputs/evidence/
```

Additional Stage I repeats use `scripts/model_training/canonical/run_5model_frozen_4source_seeds.sh` for seeds `20260612` and `20260718`; their seed-labeled evaluations use `scripts/subbench/coordinated/run_5model_frozen_4source_multi_seed_parallel.sh` and remain distinct from the main campaign. The shared Stage I/II subject manifest fixes the partition; evidence assembly instead reads the separate Stage II campaign allowlist. Cross-seed scorecards extract the same pre-specified endpoint from each seed-specific result before computing their declared mean and sample standard deviation.

Epistemic Boundaries of Reported Metrics. Appendix C reports all four task scorecards, response summaries, and the Claim–Evidence Table, while Appendix B.7 reports the clean-baseline subject-bootstrap confidence intervals. Full machine-readable condition records, Unit profiles, control curves, seed-specific Stage I and Stage II results, bootstrap artifacts, available integrity hashes, and explicit missing-hash states are preserved in the repository release. The three-seed scorecard standard deviations quantify training-state variation under the fixed partition and perturbation realizations; they do not quantify subject-level or realization-level population uncertainty.